

MAT 240 Algebra I

Brian Zu

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0.1 Introduction

The course is mainly based on the book *Linear Algebra 4th edition by S.H.Friedberg*, coupled with *A Readable Introduction to Real Mathematics* by F. Su

Chapter 1

Week 1

1.1 Sets, Functions, Fields

Definition 1.1.1: Sets

A set is a collection of elements. If x is in the set S , we write $x \in S$. If not, we write $x \notin S$. Two ways of defining a set:

1. Listing the objects in that set

$$S = \{1, 2, 3, 6\} = \{2, 1, 3, 6\}$$

Note:-

Note that the order of elements does not matter, and repeated elements are only listed once.

2. Describing or characterizing the elements of S

$$A = \{x : x \text{ is a positive integer dividing } 6\}$$

Note:-

$$A = S$$

Example 1.1.1

$$P = \{x : x \text{ is a prime number}\}$$

- $\mathbb{Z}$ integers
- $\mathbb{Q}$ the rational numbers
- $\mathbb{R}$ the real numbers

Two sets are equal if they contain the **same elements** if S and T are equal, we write $S = T$

Example 1.1.2

$$B = \{-1, 0, 1, 2, 3, 4, 5, 6, 7\}$$

$$C = \{2, 3\} = \{x : x \text{ is an integer between 2\&3}\}$$

Definition 1.1.2: Subset

If S and T are set such that all elements of s are contained in T , we say that S is a subset of T and we write $S \subset T$

If S is a subset of T and $S \neq T$, we write $S \subsetneq T$ and say that S is a proper subset of T

Definition 1.1.3: Empty Set

The empty set is the set containing no elements, denoted by $\emptyset$ or $\{\}$. If S and T are sets. $S \cup T = \{x : x \in S \text{ or } x \in T\}$ is called the **union** of S and T . $S \cap T = \{x : x \in S \text{ and } x \in T\}$ is called the intersection of S and T .

Example 1.1.3

$$N = \{x \in \mathbb{Z} : x > 0\} = \{1, 2, 3, 4, \dots\}$$

Let $-N = \{x \in \mathbb{Z} : x < 0\} = \{-1, -2, -3, -4, \dots\}$ then $N \cap -N = \emptyset$ and $N \cup -N = \{x \in \mathbb{Z} : x \neq 0\}$. If we have a list $S_1, \dots, S_k$ of sets, we can write

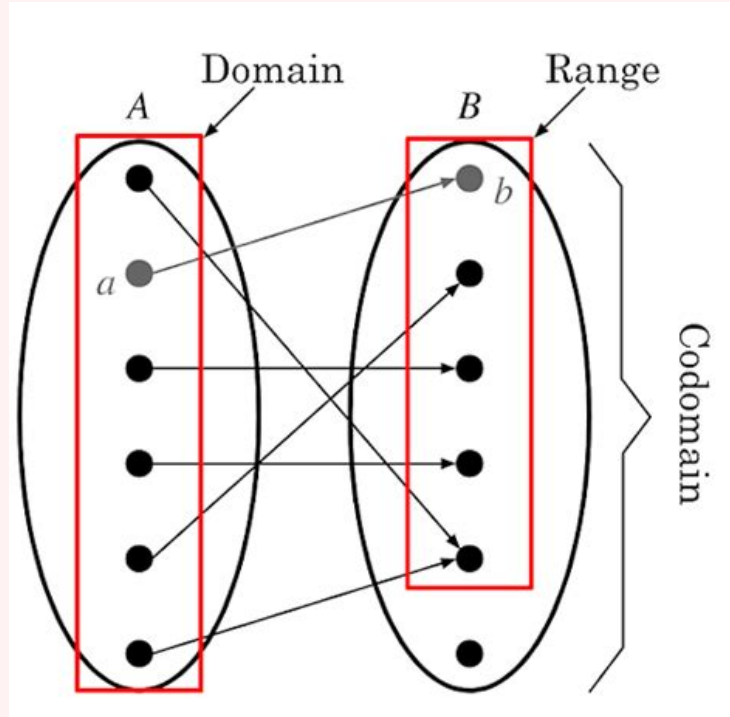
$$\bigcup_{i=1}^k S_i = \{x : x \in S_i \text{ for at least one of the } S_i, i = 1, \dots, k\}$$

$$\bigcap_{i=1}^k S_i = \{x : x \in S_i \text{ for all of the } S_i, i = 1, \dots, k\}.$$

Definition 1.1.4: Function

Consider two sets A and B , and suppose that with each element x of A there is associated, in some manner, an element of B , which denoted by $f(x)$. Then f is said to be a function from A to B , denoted by $f : A \rightarrow B$.

- The set A is called the **domain** of f
- The set B is called the **codomain** of f
- The elements $f(x)$ are called the **values** of f
- The set of all values of f is called the **range** of f



If $E \subset A$, $f(E)$ is defined to be the set of all elements $f(x)$, for $x \in E$. We call $f(E)$ the **image** of E under f .

If $E \subset B$, $f^{-1}(E)$ denotes the set of all $x \in A$ such that $f(x) \in E$. We call $f^{-1}(E)$ the **preimage** of E under f .

Definition 1.1.5: Equality of functions

Two functions $f, g : S \rightarrow T$ are equal if $f(x) = g(x)$ for all $x \in S$

Example 1.1.4

For example, let $S = \{x \in \mathbb{R} : |x| \geq 1\}$ $f : S \rightarrow \mathbb{R}$ is the function s.t. $f(x) = \frac{1}{x^2}$. Then

- domain is S
- codomain is $\mathbb{R}$
- range is $\{x \in \mathbb{R} : 0 < x < 1\} \subset \mathbb{R}$

Definition 1.1.6: Injective, surjective, and bijective

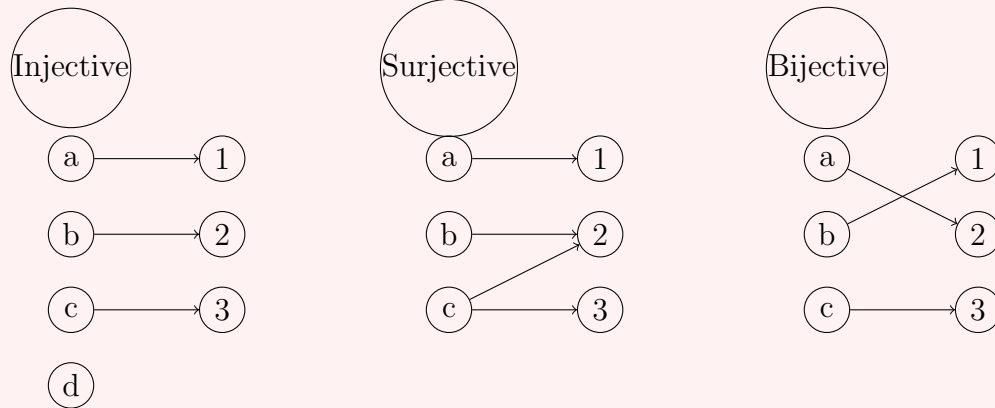
If every element in the range of f has a unique preimage, we say that f is **injective** or one-to-one. In notation:

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

A counter example would be $f(x) = x^2$. If every element in the codomain is in the image, we say that f is **surjective** (onto). In other words, the range equals the codomain. In notation:

$$\forall y \in B, \exists x \in A \text{ such that } f(x) = y$$

If f is surjective and injective, we say that F is bijective or a bijection. It's perfect pairing — each input gives a unique output, and every element of the codomain is used.



If $f : S \rightarrow T$ and $U \subset S$ the **restriction** of f to U is denoted $f|_U : U \rightarrow T$. $f|_U(u) = f(u)$ for all $u \in U$

Example 1.1.5

Let $f : \mathbf{Z} \rightarrow \mathbf{N}$ be defined by $f(x) = |x| + 1$. Then f is surjective but not injective. The function $f|_{\mathbf{N}}$ is injective but not surjective because 1 is in its range but doesn't have a preimage. Let $\mathbf{Z}_{\geq 0} = \{x \in \mathbf{Z} : x \geq 0\}$. Then $f|_{\mathbf{Z}_{\geq 0}}$ is both injective and surjective, hence a bijection.

Example 1.1.6 (Composite functions)

let S, T, U as sets,

$$f : S \rightarrow T, g : T \rightarrow U$$

The composite $g \circ f : S \rightarrow U$.

Note:-

Note that given S, T and $f, g : S \rightarrow S$. $f \circ g$ doesn't necessarily equal to $g \circ f$. For example $f(x) = x + 1$ and $g(x) = 2x$.

Example 1.1.7

Exercise: check that f, g is invertible and $s \in S$, $f^{-1}(s)$.
 Is $f : \mathbb{R} \rightarrow \mathbb{R}$ invertible?

$f(x) = 3x + 1$ has inverse $g(x) = (x - 1)/3$ Is $f : \mathbb{Z} \rightarrow \mathbb{Z}$ invertible?
 For $f(z) = 3z + 1$, the inverse $g(x) = (x - 1)/3$ will make some of the output not integers.
 For example $f(g(2))$.

Question 1

Exercise: how to prove a function is invertible if it's bijective?

Note:-

The definition of field and properties are noted down in note for baby Rudin, thus disregard.

Definition 1.1.7: Equivalence relations

- F(1) **Commutativity** - $a + b = b + a$ and $a \cdot b = b \cdot a$
- F(2) **Associativity** - $(a + b) + c = a + (b + c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- F(3) **Existence of Identity Elements** - There exists elements $0 \in F$ and $1 \in F$ such that $0 + a = a$ and $1 \cdot a = a$. 0 is called the **additive identity** and 1 is the **multiplicative identity**. Note that $0 \neq 1$
- F(4) **Existence of Inverses** - For each $a \in F$ there exists $b \in F$ and nonzero $c \in F$ such that $a + b = 0$ and $a \cdot c = 1$
- F(5) **Distributivity** - $a \cdot (b + c) = ab + ac$

Definition 1.1.8: Equivalence relations

Let S be a set. A relation on S is a subset $R \subset S \times S$. We say a pair $(x, y) \in S \times S$ satisfies the notation if $(x, y) \in R$. We typically denote the relation $\sim$ and write $x \sim y$ if $(x, y) \in R$. For example $\sim$ could be $=$

Definition 1.1.9

A relation on S is an equivalence relation if it satisfies the following 3 properties $\forall x, t \in S$

- $s \sim s$
- $s \sim t \Rightarrow t \sim s$
- $s \sim t$ and $t \sim u \Rightarrow s \sim u$

Definition 1.1.10

If $(S, \sim)$ is a set equipped with an equivalence relation and $t \in S$, let $C_t = \{s \in S : s \sim t\}$

Theorem 1.1.1

Let $(S, \sim)$ is a set equipped with an equivalence relation, then $\exists T \subset S$ such that

- $S = \bigcup_{t \in T} C_t$
- $C_t \cap C_{t'} = \emptyset$ if $t \neq t'$

We call T a set of representation for $\sim$

Chapter 2

Week 2

2.1 Modular Arithmetic

Definition 2.1.1

Let $m \in \mathbb{N}$. Let $a, b \in \mathbb{Z}$. We say that a is congruent to $b \pmod{m}$, denoted $a \equiv b \pmod{m}$. If $m \mid (b - a)$.

Example 2.1.1

Let $a = 3$, $b = 45$. Then $a \equiv b \pmod{2}$ and $\pmod{3}$

Lemma 2.1.1 $a \equiv b \pmod{m}$ if and only if (iff) there exists $k \in \mathbb{Z}$ such that $a = b + km$.

Proof: Suppose $a \equiv b \pmod{m}$ then by definition $m \mid b - a$. This means that $b - a$ is a multiple of m , so it can be written as $b - a = mk$ for some $k \in \mathbb{Z}$. Rearranging, we have $b = a + mk$ or $a = b + (-k)m$. $-k$ is our desired multiple.

From another direction, suppose that $a = b + mk$ then $b - a = -mk$ which is divisible by m . By definition, $a \equiv b \pmod{m}$. ☺

Theorem 2.1.1 Let $m \in \mathbb{N}$. For each $a \in \mathbb{Z}$, there exists a unique $r \in \{0, \dots, m - 1\}$ such that $a \equiv r \pmod{m}$.

Proof: We will prove that there exist unique integers k, r such that $a = km + r$ with $0 \leq r < m$. By a previous lemma, this implies $a \equiv r \pmod{m}$.

Existence. Consider all multiples of m , $\{0, \pm m, \pm 2m, \dots\}$, which cover the real line by the Archimedean property. Since a is an integer, there exists k such that

$$km \leq a < (k + 1)m.$$

Subtracting km gives $0 \leq a - km < m$. Define $r = a - km$, then $a = km + r$ with $0 \leq r < m$.

Uniqueness. Suppose $a = km + r = k'm + r'$ with $0 \leq r, r' < m$. Then $r - r' = (k' - k)m$, so $r - r'$ is a multiple of m . On the other hand, since both r and r' lie in $[0, m)$, their

difference satisfies

$$-m < r - r' < m.$$

The only multiple of m in this range is 0, hence $r - r' = 0 \implies r = r'$ and then $k = k'$. ☺

For the equivalence relation of congruence modulo m , we have

$$\mathbb{Z} = C_0 \cup C_1 \cup \dots \cup C_{m-1},$$

where $C_r = \{a \in \mathbb{Z} : a \equiv r \pmod{m}\}$. We call $\{0, \dots, m-1\}$ the *standard representatives*.

Theorem 2.1.2 If $a \equiv b(m)$ and $c \equiv d(m)$, then $a + c \equiv b + d(m)$, $a \cdot c \equiv b \cdot d(m)$

Proof: By lemma, $a = b + km$, $c = d + k'm$.

$$a + c = b + km + d + k'm = b + d + (k + k')m \equiv b + d(m)$$

$$a \cdot c = (b + km)(d + k'm) = bd + kmd + k'mb + kk'm^2 = bd + m(kd + k'b + kk'm) \equiv bd(m)$$

☺

Definition 2.1.2

The integers modulo m , denoted $\mathbb{Z}/m\mathbb{Z}$ is the set of equivalence classes mod m . We will denote it $\mathbb{Z}/m\mathbb{Z} = \{\bar{0}, \bar{1}, \dots, \bar{m-1}\}$. Thanks to the last theorem, $\mathbb{Z}/m\mathbb{Z}$ is equipped with addition and multiplication. Regular reputation in $\mathbb{Z}+$ remainder after division by m .

Definition 2.1.3

A prime number is $p \in \mathbb{N}$, $p > 1$, and s.t. For all $x, y \in \mathbb{Z}$, if $p \nmid x, p \nmid y \Rightarrow p \nmid xy$

Theorem 2.1.3 $\mathbb{Z}/m\mathbb{Z}$ is a field iff m is a prime

Proof: Suppose that m is **not** a prime. Then $\exists 0 < r, s < m$, such that $m = r \cdot s$. Assume for contradiction that there $\exists t, u$ s.t. $r \cdot t = 1, s \cdot u = 1$ which are the existence of multiplicative inverses. Then in $\mathbb{Z}/m\mathbb{Z}$. $1 = 1 \cdot 1 = s \cdot u \cdot r \cdot t = u \cdot t \cdot r \cdot s = u \cdot t \cdot 0 = 0$. Since $1 \neq 0$, this is a contradiction. So m is not prime $\Rightarrow \mathbb{Z}/m\mathbb{Z}$ is not a field. Assuming that m is **prime**.

Lemma 2.1.2 if p is prime and $p \nmid x, p \nmid y$, then $p \nmid xy$

Claim 2.1.1 Let $x \in \mathbb{Z}/m\mathbb{Z}$ and m is prime then if $x \neq 0$

Proof: Let $a, b \in \mathbb{Z}$, suppose $a \not\equiv b(m)$, then $m \nmid a - b$. Fix a representative of $x \in \mathbb{Z}$ call it $\tilde{x}$. Since $x \neq 0, m \nmid \tilde{x}$. Since m is prime, $m \nmid \tilde{x}(a - b)$, so $\tilde{x}a \not\equiv \tilde{x}b \pmod{m}$. I.e $xa \neq xb \in \mathbb{Z}/m\mathbb{Z}$. This proves the claim. ☺

We know multiplication by x is **injective**. By HW1, Q1(b) is also **surjective**. In other words, $\forall s \in \mathbb{Z}/m\mathbb{Z}, \exists y$ s.t. $xy = s$. In particular, $\exists y$ s.t. $xy = 1$. It follows that any element $x \neq 0$ has a multiplicative inverse, so if m is prime, $\mathbb{Z}/m\mathbb{Z}$ is a field. Upshot: $\mathbb{Z}/p\mathbb{Z}$ is a field for all prime. We will denote it $\mathbb{F}_p$ ☺

Once proved that multiplicative and additive inverses of a are unique in a field, we'll denote them respectively by a^{-1} and $-a$.

Chapter 3

Week 3

3.1 Vector Space

Definition 3.1.1: Vector Space

Let $\mathbb{F}$ be a field. A vector space over $\mathbb{F}$ is a set V equipped with two operations:

1. $+: V \times V \rightarrow V, (v, w) \mapsto v + w$
2. $\mathbb{F} \times V \rightarrow V, (a, v) \mapsto av$, where a is a scalar product of v .

Elements of $\mathbb{F}$ will be called scalars. Satisfying that for all $x, y, z \in V, a, b \in \mathbb{F}$:

1. **Commutativity of addition** $x + y = y + x$
2. **Associativity** $(x + y) + z = x + (y + z)$
3. **Additive identity** $\exists 0 \in V$ s.t. $0 + x = x \forall x \in V$
4. **Inverses** $\forall x \in V, \exists y$ s.t. $x + y = 0$
5. **Multiplicative identity** $1x = x$
6. **Associativity of multiplication** $a(bx) = (ab)x$
7. **Distributivity of scalar multiplication over vector addition** $a(x + y) = ax + ay$
8. **Distributivity of scalar addition over scalar multiplication** $(a + b)x = ax + bx$

Example 3.1.1

$$\mathbb{F}^n = \{(a_1, a_2, \dots, a_n) : a_i \in \mathbb{F} \ 1 \leq i \leq n\}$$

a_i is the i th coordinate of $(a_1, \dots, a_n)$

$$(a_1, \dots, a_n) + (b_1, \dots, b_n) = (a_1 + b_1, \dots, a_n + b_n)$$

$$c \in \mathbb{F}, \ c(a_1, \dots, a_n) = (ca_1, \dots, ca_n)$$

Example 3.1.2

Define: An $m \times n$ matrix with entries in $\mathbb{F}$ is an array of elements of $\mathbb{F}$ with m rows and n columns.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Example 3.1.3

Let S be a non empty set. Let $\mathcal{F}(S, F)$ be the set of all functions $f : S \rightarrow F$.

$$\mathcal{F}(S, F) = \{f : S \rightarrow F\}$$

Definition 3.1.2: Properties of vector space

Let V be a vector space over $\mathbb{F}$. Then

- If $x, y, z \in V$ and $x + z = y + z$ then $x = y$
- The vector $0 \in V$ is unique
- Let $x \in V$. The vector y s.t. $x + y = 0$ is unique, call it $-x$.

These properties could be derived from the axioms.

Theorem 3.1.1

Let V be a vector space over $\mathbb{F}$. Then

1. $0v = 0 \forall v \in V$
2. $(-a)v = -(av) = a(-v) \forall a \in \mathbb{F}, v \in V$
3. $0a = 0$

Note:-

Note that the proofs are in notes for baby Rudin

Proof: 1.

$$0v = (0 + 0)v = 0v + 0v$$

By the cancellation property, $0v$ on both sides can be canceled, so $0 = 0v$

2.

$$\begin{aligned} (-a)v + av &= v(-a + a) = 0v = 0 \\ \Rightarrow (-a)v &= av \end{aligned}$$

By doing it the other way round, we get $av = a(-v)$

3.

$$0a + 0a = (0 + 0)a = 0a$$

Since given $x + y = x \Rightarrow y = 0$, $0a = 0$



Definition 3.1.3: x-y

Now we can define $x - y$ by $x + (-y)$

Definition 3.1.4: Subspaces

Let V be a vector space over $\mathbb{F}$. We say that $W \subset V$ is a **subspace** of V if it is a vector space over $\mathbb{F}$ under the restriction of the operations of V .

Since the operations on V satisfy the properties above, it is only needed to check a few things.

Theorem 3.1.2

Let V be a vector space. Then $W \subset V$ is a subspace **iff** the following hold.

1. $W \neq \emptyset$
2. $\forall x, y \in W, x + y \in W$ (closed under addition)
3. $c \in \mathbb{F}, x \in W, \Rightarrow cx \in W$ (closed under scalar multiplication)

Proof: (Direction $\Leftarrow$) Assume that W satisfies (2) and (3). Then the operations $+$ and $\cdot$ on W are well-defined.

Axioms 1,2,5,6,7,8 all automatically hold since the operation on W are defined from those on V (inherited from). In other words, **additive identity and inverses** are the axioms to be checked.

Additive identity By (1) $\exists w \in W$. Then $0w = 0$ so W has a 0.

Inverses Let $w \in W$. Then $(-1)w \in W$ by (3).

$$(-1)w + w = (-1 + 1)w = 0w = 0$$

So $(-1)w$ is the additive inverse of w and is contained in W .

(Direction $\Rightarrow$) Suppose W is a subspace. Then $0 \in W$ and by definition $\forall x, y \in W, x + y \in W$ and $\forall c \in \mathbb{F}$ and $x \in W, cx \in W$. ☺

Corollary 3.1.1

If $W \subset V$ is a subspace, the $0w = 0v$ and in particular $0v \in W$.

Example 3.1.4

$$\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$$

$$W = \{(x, x) : x \in \mathbb{R}\}$$

Theorem 3.1.3

The intersection of two subspaces is a subspace.

Proof: Let W_1, W_2 be subspace of V

- $0 \in W_1 \cap W_2$
- If $x, y \in W_1 \cap W_2$ then

$$x + y \in W_1 \quad \text{since } W_1 \text{ is a subspace}$$

$x + y \in W_2$ since W_2 is a subspace

$\Rightarrow x + y \in W_1 \cap W_2$.

- if $x \in W_1 \cap W_2$, $c \in \mathbb{F}$

$cx \in W_1$ since W_1 is a subspace

$cx \in W_2$ since W_2 is a subspace

$\Rightarrow cx \in W_1 \cap W_2 \Rightarrow W_1 \cap W_2$ is a field.



3.2 Linear combination

Definition 3.2.1

Let V be a vector space and $S \subseteq V, S \neq \emptyset$. A vector $v \in V$ is a linear combination of elements of S if there exists $u_1, \dots, u_n \in S, d_1, \dots, d_n \in F$ such that $v = a_1u_1 + \dots + a_nu_n$. We say that v is a linear combination of $u_1, \dots, u_n$. The a are called the coefficients.

Example 3.2.1

Let $V = P(Q) = \{a_1x, \dots, a_nx^n : a_x \in \mathbb{Q}\}$ be a vector space. Then $3x^2 + 1$ is a linear combination of x and 1 . The coefficients are $3, 1$.

3.3 Span

Definition 3.3.1

Let S be a non empty set be a subset of V . The span of S , denoted $\text{span}(S)$ is the set of all linear combinations of elements of S . If $\text{span}(S) = V$, we say that S spans V .

Example 3.3.1

Let $V = \mathbb{R}^3$ what's the span for $(1, 0, 1), (0, 0, 3)$

$$a, 0, c, a, c \in \mathbb{R}$$

Theorem 3.3.1

The span of any $S \subset V$ is a subspace W such that $S \subset W$. Moreover, any subspace containing S will also contain W .

Proof: We will check the three properties of subspace:

- $\text{span}(S)$ is not empty by definition.
- If $u = \sum_{i=1}^n a_i u_i, w = \sum_{j=1}^m c_j u_j$ for $u_i, u_j \in S$, then $u + w = \sum_{i=1}^n a_i u_i + \sum_{j=1}^m c_j u_j$ which is a linear combination of elements of S .

- If $c \in F$ and $v = \sum_{i=1}^n a_i u_i$ with $u_i \in S$, then $cv = \sum_{i=1}^n ca_i u_i$ which is a linear combination of elements of S so $cv \in \text{span}(S)$. $S \subset \text{span}(S)$



3.4 Linear Independence/Dependence

Definition 3.4.1: Linear relation

Let V be a vector space over $\mathbb{F}$. A subset $S \subset V$ is linearly dependent if there exists $s_1, \dots, s_n \in S$ and $c_1, \dots, c_n \in \mathbb{F}$, **not all zero** such that

$$c_1 s_1 + \dots + c_n s_n = 0$$

We call such an expression a **linear relation**. We can always do this with $c_1 = c_2 = \dots = c_n = 0$. This is called the **trivial linear relation**.

Definition 3.4.2: Linear Independence

A list $v_1, \dots, v_m$ of vectors in V is **linearly independent** if the only choice of $a_1, \dots, a_m \in \mathbb{F}$ that makes $a_1 v_1 + \dots + a_m v_m = 0$ is $a_1 = \dots = a_m = 0$. In other words, it's linearly independent if there is no nontrivial linear relation among two vectors.

Theorem 3.4.1

Let $S_1 \subset S_2 \subset V$. If S_1 is linearly dependent, then S_2 is linearly dependent.

Proof: This is in exercise



Corollary 3.4.1

Let $S_1 \subset S_2 \subset V$. If S_2 is linearly independent, then S_1 is linearly independent.

Theorem 3.4.2

Let $S \subset V$ be linearly independent, and let $v \in V \setminus S$. Then $S \cup \{v\}$ is linearly independent $\iff v \notin \text{span}(S)$.

Proof: ($\Rightarrow$) Suppose $S \cup \{v\}$ is linearly independent. Assume for contradiction that $v \in \text{span}(S)$. Then there exist scalars $c_1, \dots, c_n \in \mathbb{F}$ and $s_1, \dots, s_n \in S$ such that

$$v = \sum_{i=1}^n c_i s_i.$$

Rearranging gives

$$1 \cdot v + \sum_{i=1}^n (-c_i) s_i = 0.$$

This is a nontrivial linear relation among elements of $S \cup \{v\}$, contradicting linear independence. Hence $v \notin \text{span}(S)$.

($\Leftarrow$) Suppose $v \notin \text{span}(S)$. We show $S \cup \{v\}$ is linearly independent. Consider a linear relation

$$av + \sum_{i=1}^n c_i s_i = 0$$

with $s_i \in S$, $a, c_i \in \mathbb{F}$. If $a = 0$, then we have $\sum_{i=1}^n c_i s_i = 0$. Since S is linearly independent, this implies all $c_i = 0$, so the relation is trivial.

If $a \neq 0$, then we can solve for v :

$$v = \sum_{i=1}^n \left(-\frac{c_i}{a}\right) s_i,$$

which shows $v \in \text{span}(S)$, a contradiction. Hence the case $a \neq 0$ cannot occur, so the only possible relation is trivial.

Therefore $S \cup \{v\}$ is linearly independent. $\odot$

Chapter 4

Week 4

4.1 Bases

Definition 4.1.1

A basis b for a vector space V is a subset $b \subset V$ such that:

1. B is linearly independent
2. B spans V

If b is a basis for V , we say that the vectors in b form a basis of V .

Theorem 4.1.1

Let V be a vector space, and $u_1, \dots, u_n$ be distinct vectors in V . Then $b = \{u_1, \dots, u_n\}$ is a basis of V if and only if each $v \in V$ can be written uniquely as a linear combination of the vectors in b , i.e.

$$v = c_1 u_1 + \dots + c_n u_n$$

for unique scalars $c_1, \dots, c_n$.

Proof: ($\Rightarrow$) Suppose b is a basis of V .

Since b spans V , every $v \in V$ can be written as

$$v = c_1 u_1 + \dots + c_n u_n$$

for some scalars $c_1, \dots, c_n$. This proves *existence*.

For *uniqueness*, suppose

$$v = c_1 u_1 + \dots + c_n u_n = d_1 u_1 + \dots + d_n u_n.$$

Subtracting, we obtain

$$0 = (c_1 - d_1)u_1 + \dots + (c_n - d_n)u_n.$$

Since b is linearly independent, each coefficient must vanish, so $c_i = d_i$ for all i . Thus the representation is unique.

($\Leftarrow$) Conversely, assume that every $v \in V$ can be written uniquely as a linear combination of the vectors in b .

Since every vector has such a representation, b spans V .

To prove linear independence, suppose

$$0 = c_1 u_1 + \cdots + c_n u_n.$$

But also

$$0 = 0 \cdot u_1 + \cdots + 0 \cdot u_n.$$

By uniqueness of representation, we must have $c_i = 0$ for all i . If there are some other nontrivial linear relation such that $0 = c_1 u_1 + \cdots + c_n u_n$, where not all $c_i = 0$, then it contradicts with the uniqueness assumption. Thus b is linearly independent.

Therefore b is a basis of V . ☺

Theorem 4.1.2

If V is a vector space with a finite spanning set, then V has a finite basis.

Proof: Let S be a finite spanning set of V . Our goal is to reduce S step by step until we obtain a basis.

Step 1: Check if S is linearly independent.

If S is already linearly independent, then S itself is a finite basis of V , and we are done.

Step 2: Remove one redundant vector if S is dependent.

Otherwise, S is linearly dependent. Thus there exist distinct vectors $v_1, \dots, v_n \in S$ and scalars $a_1, \dots, a_n \in \mathbb{F}$, not all zero, such that

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n = 0.$$

Choose an index i with $a_i \neq 0$. Then we can solve for v_i :

$$v_i = (-a_i)^{-1} \sum_{k \neq i} a_k v_k.$$

This shows that v_i is a linear combination of the other vectors in S .

Step 3: Show that the span does not change.

Let $S' = S \setminus \{v_i\}$. We claim that

$$\text{span}(S) = \text{span}(S').$$

Indeed, since $S' \subseteq S$, it follows immediately that $\text{span}(S') \subseteq \text{span}(S)$. Conversely, take any $u \in \text{span}(S)$. Then

$$u = c v_i + \sum_{j=1}^r c_j u_j, \quad \text{for some } u_j \in S.$$

Substituting the expression for v_i in terms of the other vectors, we obtain

$$u = c(-a_i)^{-1} \sum_{k \neq i} a_k v_k + \sum_{j=1}^r c_j u_j,$$

which is a linear combination of elements of S' . Thus $u \in \text{span}(S')$. Since $u \in \text{span}(S)$ was arbitrary, we conclude $\text{span}(S) \subseteq \text{span}(S')$. Therefore $\text{span}(S) = \text{span}(S')$.

Step 4: Repeat the process.

We have shown how to remove one redundant vector while preserving the span. If S' is linearly independent, then S' is a basis and we are finished. If S' is still dependent, we can repeat the process, removing one vector at a time while keeping the same span.

Step 5: Termination.

Since S is finite, this process must terminate after finitely many steps. The result is a subset $B \subseteq S$ which is finite, linearly independent, and still spans V . Thus B is a finite basis of V .

Special case. If $V = \{0\}$, then the empty set $\emptyset$ is by convention a basis, and it is finite. ☺

Theorem 4.1.3

Suppose V has a spanning set G consisting of n elements. Let $L \subset V$ be finite, linearly independent, with m elements. Then:

1. $m \leq n$.
2. There exists a subset $H \subset G$ of size $n - m$ such that $L \cup H$ spans V .

Proof: We argue by induction on $m = |L|$.

Base case: If $m = 0$, then $L = \emptyset$. Clearly $0 \leq n$, and choosing $H = G$ (of size n) gives $L \cup H = G$, which spans V .

Inductive step: Assume the result holds for all linearly independent sets of size m . Now let $L = \{v_1, \dots, v_{m+1}\}$ be linearly independent. Then $\{v_1, \dots, v_m\}$ is linearly independent, so by the inductive hypothesis, there exist $u_1, \dots, u_{n-m} \in G$ such that

$$\{v_1, \dots, v_m\} \cup \{u_1, \dots, u_{n-m}\}$$

spans V .

In particular, v_{m+1} can be expressed as

$$v_{m+1} = a_1 v_1 + \dots + a_m v_m + b_1 u_1 + \dots + b_{n-m} u_{n-m}.$$

Since L is linearly independent, not all b_j can vanish. Without loss of generality, suppose $b_1 \neq 0$. Then we may solve for u_1 :

$$u_1 = b_1^{-1} \left(v_{m+1} - (a_1 v_1 + \dots + a_m v_m + b_2 u_2 + \dots + b_{n-m} u_{n-m}) \right).$$

Hence $u_1 \in \text{span}(\{v_1, \dots, v_{m+1}, u_2, \dots, u_{n-m}\})$.

It follows that

$$V = \text{span}(\{v_1, \dots, v_m, u_1, \dots, u_{n-m}\}) = \text{span}(\{v_1, \dots, v_{m+1}, u_2, \dots, u_{n-m}\}).$$

Therefore, if we let $H = \{u_2, \dots, u_{n-m}\}$, which has size $n - (m + 1)$, we obtain $L \cup H$ as a spanning set.

Thus the theorem holds for $m + 1$. By induction, it holds for all m . $\odot$

Corollary 4.1.1

If V has a finite basis, then all bases of V are finite and have the same cardinality.

Proof: Let B, B' be two bases of V . Since B' spans V and B is linearly independent, the theorem gives $|B| \leq |B'|$. Reversing the roles gives $|B'| \leq |B|$. Hence $|B| = |B'|$. Finally, if V had one finite basis B and one infinite basis A , then picking $|B| + 1$ elements of A would give a linearly independent set larger than B , contradicting the theorem. Thus all bases have the same finite cardinality, called the dimension of V . $\odot$

Chapter 5

Week 5

5.1 Dimension

Definition 5.1.1: Dimension

A vector space is finite dimensional if it has a finite basis b . The unique integer n such that $|b| = n$ is called the dimension of V , denoted $\dim(V)$

Definition 5.1.2: Finite dimensional vector space

A vector space is called **finite-dimensional** if some list of vectors in it spans the space

Definition 5.1.3: Infinite dimensional vector space

A vector space that is not finite dimensional is infinite dimensional

Example 5.1.1 (Infinite dimensional vector space)

- $P(F)$, a polynomial of any degree
- $F(\mathbb{R}, \mathbb{R})$
- $\mathbb{R}$ over $\mathbb{Q}$

Theorem 5.1.1

If a vector space V is generated by a finite set S , then some subset of S is a basis for V . Hence V has a finite basis.

Corollary 5.1.1 of the replacement theorem

Let V be a vector space of dimension n .

1. Any finite generating (or spanning) set for V contains at least n elements. Any generating set that contains exactly n elements is a basis.

Proof: Let G be a finite generating set for V .

☺

2. Any linearly independent set that contains exactly n vectors is a basis.

3. Every linearly independent sets can be extended to a basis. I.e. if L is linearly independent, there exists a basis b of V such that $L \subset B$.

Definition 5.1.4: Dimension of subspace

Let V be a finite dimensional vector space, and let W be a subspace. Then W is finite dimensional, and $\dim(W) \leq \dim(V)$. Moreover, if $\dim(W) = \dim(V) \Rightarrow W = V$.

Proof: Let $\dim(V) = n$. If $W = \{0\}$, then W is finite dimensional and $\dim(W) = 0 \leq n$. Otherwise, W contains a nonzero vector x_1 ; so $\{x_1\}$ is a linearly independent set. Continue choosing vectors, $x_1, x_2, \dots, x_k$ in W such that $\{x_1, x_2, \dots, x_k\}$ is linearly independent. Since no linearly independent subset of V can contain more than n vectors, this process must stop at a stage where $k \leq n$ and $\{x_1, x_2, \dots, x_k\}$ is linearly independent but adjoining any other vectors from W produces a linearly dependent set. By 3.4, it implies that $\{x_1, x_2, \dots, x_k\}$ spans W , and hence it is a basis for W . Therefore $\dim(W) = k \leq n$.

If $\dim(W) = n$, then a basis for W is a linearly independent subset of V containing n vectors. But by the corollary of the replacement theorem 5.1 implies that this basis for W is also a basis for V ; so $W = V$. ☺

Question 1: Thinking question

Let V be a 3 dimensional vector, let W_1, W_2 be a 2 dimensional subspace. Show that $W_1 \cap W_2 \neq \{0\}$.

Note:-

This will be proven during tutorial

Theorem 5.1.2 Vector space has basis

Every vector space has a basis.

5.2 Maximal Linearly Independent Sets

Definition 5.2.1

Let F be a family of sets. A member M of F is called maximal (with respect to inclusion) if M is contained in no member of F other than M itself.

Definition 5.2.2

Let S be a subset of vector space V . A **maximal linearly independent subset** of S is a subset B of S satisfying both of the following conditions:

- B is linearly independent
- The only linearly independent subset of S that contains B is B itself.

Theorem 5.2.1

If B is a maximal linearly independent subset, then it is a basis of V .

Theorem 5.2.2

Every vector space contains a maximal linearly independent subset.

Chapter 6

Week 6

6.1 Linear transformation

Definition 6.1.1: Linear transformation

Let V, W be vector spaces over F . A **linear transformation** from V to W is a function

$$T : V \rightarrow W$$

such that for all $x, y \in V$, all $c \in F$

- $T(x + y) = T(x) + T(y)$
- $T(cx) = cT(x)$

In that case, we can say that T is linear.

Example 6.1.1

Check the following:

- **Addition:** $T(0) = 0$
- **Multiplication:** T is a linear iff for any $x_1, \dots, x_n \in V$, $c_1, \dots, c_n \in F$ $T(c_1x_1 + \dots + c_nx_n) = c_1T(x_1) + \dots + c_nT(x_2)$

Example 6.1.2

$\mathbb{R}_x : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. $\mathbb{R}_x(a, b) = (a, -b)$. This function reflects a vector by x axis. We now want to check if this is linear transformation.

- $\mathbb{R}_x((a_1, b_1) + (a_2, b_2)) = \mathbb{R}_x((a_1 + a_2, b_1 + b_2)) = (a_1 + a_2, -b_1 - b_2) = (a_1, -b_1) + (a_2, -b_2) = \mathbb{R}_x(a_1, b_1) + \mathbb{R}_x(a_2, b_2)$
- $\mathbb{R}_x(c(a, b)) = \mathbb{R}_x((ca, cb)) = (ca, -cb) = c(a, -b) = c\mathbb{R}_x(a, b)$

Example 6.1.3

$ev_1 : P_n(F) \rightarrow F$. $ev_1(P(x)) = P(1)$.

- Check addition, let $p(x) = a_0 + a_1x + \dots + a_nx^n$, $q(x) = b_0 + b_1x + \dots + b_nx^n$.

$$\begin{aligned}
 ev_1(p(x) + q(x)) &= ev_1(a_0 + a_1x + \dots + a_nx^n + b_0 + b_1x + \dots + b_nx^n) \\
 &= ev_1(a_0 + b_0 + (a_1 + b_1)x + \dots + (a_n + b_n)x^n) \\
 &= a_0 + b_0 + a_1 + b_1 + \dots + a_n + b_n \\
 &= a_0 + a_1 + \dots + a_n + b_0 + b_1 + \dots + b_n \\
 &= ev_1(p(x)) + ev_2(q(x))
 \end{aligned}$$

- Check scalar multiplication,

$$\begin{aligned}
 ev_1(c(p(x))) &= ev_1(ca_0 + ca_1x + \dots + ca_nx^n) \\
 &= ca_0 + ca_1 + \dots + ca_n \\
 &= c(a_0 + \dots + a_n) \\
 &= c \cdot ev_1(p(x))
 \end{aligned}$$

Definition 6.1.2: Kernel

Let V, W be vector spaces and $T : V \rightarrow W$ be linear. The **kernel** of T , $ker(T) = \{v \in V | T(v) = 0\} \subset V$. The image of T is $Im(T) = \{T(v) | v \in V\} \subset W$

Example 6.1.4 (Two important linear transformation)

$$\begin{aligned}
 id : V &\rightarrow V, id(v) = v, ker(id) = \{0\}, im(id) = V \\
 o : V &\rightarrow V, o(v) = 0, ker(o) = V, im(o) = \{0\}
 \end{aligned}$$

Theorem 6.1.1

Let V and W be vector spaces and $T : V \rightarrow W$ be linear. Then $null(T)$ and $im(T)$ are subspaces of V and W , respectively

Proof: In order to prove something is a subspace, we'll prove by showing that it satisfies the three properties. To clarify the notation, let $0_v, 0_w$ be the zero vectors of V and W , respectively.

- **Zeros** Since $T(0_v) = 0_w$, we have that $0_v \in null(T)$.
- **Closure under addition** Let $x, y \in null(T)$, then $T(x + y) = T(x) + T(y) = 0_w + 0_w = 0_w$, hence $x + y \in null(T)$
- **Closure under scalar multiplication** $T(cx) = cT(x) = c0_w = 0_w$, thus $cx \in null(T)$

So, $null(T)$ is a subspace of V .

- **Zeros** $T(0_v) = 0_w$, we have $0_w \in im(T)$.
- **Closure under addition** Now let $x, y \in im(T)$, $c \in \mathbb{F}$. Then $\exists v, w \in V$ such that $T(v) = x$ and $T(w) = y$. So $T(v + w) = T(v) + T(w) = x + y$. Thus $x + y \in im(T)$
- **Closure under scalar multiplication** $T(cv) = cT(v) = cx$. Thus $cx \in im(T)$

So, $im(T)$ is a subspace of W . ☺

Theorem 6.1.2

Let V and W be vector spaces and $T : V \rightarrow W$ be *linear*. Let b be a basis of V . Then

$$T(b) = \{T(v) : v \in b\} \subset W$$

spans $im(T)$

Proof: We will show that $span(T(b)) = im(T)$

- $(\subset)$ $T(b) = \{T(v) : v \in b\} \subset im(T)$ by definition, since $im(T)$ is a subspace,

$$T(b) \subset im(T) \Rightarrow span(T(b)) \subset im(T)$$

- $(\supset)$ Let $v \in im(T)$. Then $v = T(x)$ for $x \in V$. Since b is a basis of V , there exists $x_1, \dots, x_n \in b$, $c_1, \dots, c_n \in F$ such that $x = c_1x_1 + \dots + c_nx_n$. Since T is linear, $T(x) = c_1T(x_1) + \dots + c_nT(x_n) = v$ so $v \in span(T(b)) \Rightarrow im(T) \subset span(T(b))$

☺

Definition 6.1.3: Nullity and Rank

Let V, W be vector spaces and let $T : V \rightarrow W$ be linear. If $ker(T)$ and $im(T)$ are finite dimensional, we call their dimensions the nullity $\emptyset$ the rank of T , respectively. We denote them $dim(ker(T)) = null(T)$, $dim(im(T)) = rank(T)$

Theorem 6.1.3 Dimension theorem/Rank-Nullity theorem

Note:-

Prof's favourite theorem

Let V and W be vector spaces with V being finite dimensional. Let $T : V \rightarrow W$ be linear, then

$$dim(V) = rank(T) + null(T)$$

Proof: Let $dim(V) = n$ and $b = \{v_1, \dots, v_n\}$ be the basis of V . Let $\{u_1, \dots, u_k\}$ be the basis of $ker(T)$, so in particular $null(T) = k \leq n$. By the replacement theorem, exists a subset of b' consisting of $n - k$ elements, without loss of generality (wlog), $\{v_{k+1}, \dots, v_n\}$ such that $b = \{u_1, \dots, u_k\} \cup \{v_{k+1}, \dots, v_n\}$ is a basis of V . By the previous theorem 6.1, $im(T) = span(T(b)) = span(T(b')) = span(\{T(v_{k+1}), \dots, T(v_n)\})$ since $T(v_i) = 0$. We need to show that

$$\{T(v_{k+1}), \dots, T(v_n)\}$$

is linearly independent. Assume that $c_{k+1}, \dots, c_n \in \mathbb{F}$ such that $c_{k+1}T(v_{k+1}) + \dots + c_nT(v_n) = 0$.

$$T(c_{k+1}v_{k+1} + \dots + c_nv_n) = 0 \Rightarrow c_{k+1}v_{k+1} + \dots + c_nv_n \in ker(T)$$

so $\exists a_1, \dots, a_k$ such that $a_1v_1 + \dots + a_kv_k = c_{k+1}v_{k+1} + \dots + c_nv_n$ or, equivalently $a_1v_1 + \dots + a_kv_k - c_{k+1}v_{k+1} - \dots - c_nv_n = 0$. This is a linear relation between elements of b . Since b is linearly independent, $a_1 = \dots = a_k = c_{k+1} = \dots = c_n = 0$. This shows that $\{T(v_{k+1}), \dots, T(v_n)\}$ is linearly independent, so it is a basis of $im(T)$. $dim(im(T)) = n - k$

$$dim(ker(T)) + dim(im(T)) = k + n - k = n = dim(V)$$



Theorem 6.1.4

Let $T : V \rightarrow W$ be a linear transformation. Then T is injective iff $\ker(T) = \{0\}$.

Proof: proof by two directions

- $(\Rightarrow)$ Assume T is injective, let $x \in \ker(T)$. Then $T(x) = 0$. But since $T(0) = 0$, injectivity forces $x = 0$.
- $(\Leftarrow)$. Assume $\ker(T) = \{0\}$, and let x, y be such that $T(x) = T(y)$. Then $T(x - y) = 0$, so $x - y \in \ker(T)$, so $x - y = 0 \Rightarrow x = y$.



Question 1

Prove the following theorem. Let V and W be finite-dimensional vector spaces of equal dimension, and let $T : V \rightarrow W$ be linear then the following are equivalent.

- T is injective
- T is surjective
- $\text{rank}(T) = \dim(V)$

Theorem 6.1.5

Let V, W , be vector spaces over $\mathbb{F}$, and suppose that $\beta = \{v_1, \dots, v_n\}$ is a basis of V . Let $\gamma = \{w_1, \dots, w_n\} \subset W$. There exists exactly one linear transformation

$$T : V \rightarrow W \quad \text{s.t.} \quad T(v_i) = w_i, \quad 1 \leq i \leq n$$

Proof: Existence. Since $\beta = \{v_1, \dots, v_n\}$ is a basis, each $v \in V$ can be written as a linear combination $v = a_1v_1 + \dots + a_nv_n$. Define

$$T(v) = a_1w_1 + \dots + a_nw_n$$

- T is linear. If $v' = b_1v_1 + \dots + b_nv_n$, $c \in \mathbb{F}$.

$$\begin{aligned}
T(cv + v') &= T(c(a_1v_1 + \dots + a_nv_n) + b_1v_1 + \dots + b_nv_n) \\
&= T(ca_1v_1 + \dots + ca_nv_n + b_1v_1 + \dots + b_nv_n) \\
&= T((ca_1 + b_1)v_1 + \dots + (ca_n + b_n)v_n) \\
&= (ca_1 + b_1)w_1 + \dots + (ca_n + b_n)w_n \\
&= c(a_1w_1 + \dots + a_nw_n) + b_1w_1 + \dots + b_nw_n \\
&= cT(v) + T(v')
\end{aligned}$$

Clearly $T(v_i) = w_i$

Uniqueness Suppose that T' is linear and $T'(v_i) = w_i$, then $T'(a_1v_1 + \dots + a_nv_n) = a_1w_1 + \dots + a_nw_n$. So $T' = T$ and T is unique. 

Corollary 6.1.1

If β is a finite basis for V and $T, U : V \rightarrow W$ are linear with $T(v) = U(v)$ for all $v \in \beta$, then $T(v) = U(v), \forall v \in V$.

6.1.1 Matrices and Linear Transformation

Henceforth, all vector spaces are finite-dimensional.

Definition 6.1.4: Ordered basis

An ordered basis of V is a basis of V endowed with a specific order, i.e. a finite list of linearly independent vectors in V , that spans V . E.g.

$$\beta = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}, \quad \beta' = \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$

They are equal as bases but not as ordered bases.

Definition 6.1.5

Let $\beta = \{u_1, \dots, u_n\}$ be an ordered basis of V . For $v \in V$, let $a_1, \dots, a_n$ be the unique scalars such that $v = a_1u_1 + \dots + a_nu_n$. Then the coordinate vector of v with respect to β is

$$[v]_\beta = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$$

Note: $[u_i]_\beta = e_i$

Definition 6.1.6

Let V, W be finite-dimensional. Let $\beta = \{v_1, \dots, v_n\}$ be a basis of V , and $\gamma = \{w_1, \dots, w_m\}$ be a basis of W . Let $T : V \rightarrow W$ be a linear transformation. Then for each $1 \leq j \leq n$ there exists scalars $1 \leq i \leq m$ s.t.

$$T(v_j) = \sum_i^m a_{ij} w_i$$

We define the $n \times m$ matrix A given by $A_{ij} = a_{ij}$ to be the matrix representation of T in the ordered bases β and γ , and we write $A = [T]_\beta^\gamma$. If $V = W$ and $\beta = \gamma$, we write $A = [T]_\beta$.

Notice that the j th column of A is simply $[T(v_j)]_\gamma$.

Definition 6.1.7: Addition and Scalar multiplication for linear transformations

Let $T, U : V \rightarrow W$ be arbitrary functions, where V, W are vector spaces over $\mathbb{F}$, and let $a \in \mathbb{F}$. We define

- $T + U : V \rightarrow W$ by $(T + U)(x) = T(x) + U(x), \forall x \in V$
- $aT : V \rightarrow W$ by $(aT)(x) = aT(x), \forall x \in V$

Theorem 6.1.6

Let $L(V, W)$ be the set of linear transformations $T : V \rightarrow W$. $L(V, W)$ is a subspace of $F(V, W)$, where F denotes the set of all functions.

Proof: Proof is in HW6 Q2 a):

- **Zero** Since U is a subspace, $0_V \in U$. Since T is linear, $T(0_V) = 0 \cdot T(v) = 0_W$.
- **Closed under addition** Let $y_1, y_2 \in T(U)$, $y_1 = T(u_1)$, $y_2 = T(u_2)$ for some $u_1, u_2 \in U$. Since U is a subspace, $u_1 + u_2 \in U$. Then linearity of T gives

$$y_1 + y_2 = T(u_1) + T(u_2) = T(u_1 + u_2) \in T(U)$$

- **Closed under multiplication** For $\lambda \in F$, $y = T(u) \in T(U)$, since U is a subspace, $\lambda u \in U$. Linearity gives

$$\lambda y = \lambda T(u) = T(\lambda u) \in T(U)$$

Thus $T(U)$ is a subspace of W . ☺

Theorem 6.1.7

The function $[\cdot]_\beta^\gamma$ taking a linear transformation to its matrix is itself a linear transformation. In particular, for $T, U \in L(V, W)$ and $c \in \mathbb{F}$,

$$[T + U]_\beta^\gamma = [T]_\beta^\gamma + [U]_\beta^\gamma, \quad [cT]_\beta^\gamma = c [T]_\beta^\gamma.$$

Proof: Addition: Let $\beta = \{v_1, \dots, v_n\}$ be a basis of V and $\gamma = \{w_1, \dots, w_m\}$ a basis of W . For each j there exist unique scalars $a_{1j}, \dots, a_{mj}$ and $b_{1j}, \dots, b_{mj}$ such that

$$T(v_j) = \sum_{i=1}^m a_{ij} w_i, \quad U(v_j) = \sum_{i=1}^m b_{ij} w_i.$$

Then

$$(T + U)(v_j) = T(v_j) + U(v_j) = \sum_{i=1}^m (a_{ij} + b_{ij}) w_i,$$

so the (i, j) -entry of $[T + U]_\beta^\gamma$ is $a_{ij} + b_{ij}$. Since matrix addition is entrywise, $([T]_\beta^\gamma)_{ij} + ([U]_\beta^\gamma)_{ij} = a_{ij} + b_{ij}$, hence $[T + U]_\beta^\gamma = [T]_\beta^\gamma + [U]_\beta^\gamma$.

Scalar multiplication, let $c \in \mathbb{F}$. Then

$$(cT)(v_j) = cT(v_j) = c \sum_{i=1}^m a_{ij} w_i = \sum_{i=1}^m (ca_{ij}) w_i.$$

Thus the (i, j) -entry of $[cT]_\beta^\gamma$ is ca_{ij} , i.e.

$$([cT]_\beta^\gamma)_{ij} = c a_{ij} = c ([T]_\beta^\gamma)_{ij},$$

which means $[cT]_\beta^\gamma = c [T]_\beta^\gamma$.

Therefore $[\cdot]_\beta^\gamma : L(V, W) \rightarrow M_{m \times n}(\mathbb{F})$ is linear. ☺

Theorem 6.1.8

Let V, W, Z be vector spaces, let $T : V \rightarrow W$ be linear, $U : W \rightarrow Z$ be linear. Then $U \circ T : V \rightarrow Z$ is linear.

Proof: Let $v_1, v_2 \in V$ and $c \in \mathbb{F}$. For scalar multiplication:

$$(U \circ T)(cv) = U(T(cv)) = U(cT(v)) = cU(T(v)) = c(U \circ T)(v).$$

For addition:

$$\begin{aligned} (U \circ T)(v_1 + v_2) &= U(T(v_1 + v_2)) \\ &= U(T(v_1) + T(v_2)) \quad (\text{linearity of } T) \\ &= U(T(v_1)) + U(T(v_2)) \quad (\text{linearity of } U) \\ &= (U \circ T)(v_1) + (U \circ T)(v_2). \end{aligned}$$

Hence $U \circ T$ preserves both addition and scalar multiplication, so it is linear. ☺

Definition 6.1.8

$$U_1, U_2, T \in L(V, V) := L(V)$$

- $T \circ (U_1 + U_2) = T \circ U_1 + T \circ U_2, (U_1 + U_2) \circ T = U_1 \circ T + U_2 \circ T$
- $(T \circ U_1) \circ U_2 = T \circ (U_1 \circ U_2)$
- $T \circ id = id \circ T = T$

Theorem 6.1.9 Uniqueness of linear transformation

Let V and W be vector spaces over F , and suppose that $\{v_1, v_2, \dots, v_n\}$ is a basis of V . For $w_1, w_2, \dots, w_n$ in W , there exists exactly one linear transformation $T : V \rightarrow W$ such that $T(v_i) = w_i$ for $i = 1, 2, \dots, n$

Proof: Let $x \in V$, then

$$x = \sum_{i=1}^n a_i v_i$$

where $a_1, a_2, \dots, a_n$ are unique scalars. Define

$$T : V \rightarrow W \text{ by } T(x) = \sum_{i=1}^n a_i w_i$$

Linearity: Suppose that $u, v \in V$ and $d \in F$. Then we may write

$$u = \sum_{i=1}^n b_i v_i, \quad v = \sum_{i=1}^n c_i v_i$$

for some scalars $b_1, b_2, \dots, b_n, c_1, c_2, \dots, c_n$. Thus

$$du + v = \sum_{i=1}^n (db_i + c_i) v_i$$

so

$$T(du + v) = \sum_{i=1}^n (db_i + c_i)w_i = d \sum_{i=1}^n b_i w_i + \sum_{i=1}^n c_i w_i = dT(u) + T(v)$$

Clearly

$$T(v_i) = w_i, \quad i = 1, 2, \dots, n$$

Uniqueness Suppose that $U : V \rightarrow W$ is linear and $U(v_i) = w_i$ for $i = 1, 2, \dots, n$. Then for $x \in V$ with

$$x = \sum_{i=1}^n a_i v_i$$

we have

$$U(x) = \sum_{i=1}^n a_i U(v_i) = \sum_{i=1}^n a_i w_i = T(x)$$

Hence $U = T$.



6.1.2 Matrix-Vector multiplication

Definition 6.1.9

Let $A \in M_{m \times n}(\mathbb{F})$, $v \in \mathbb{F}^n$. write

$$A = [c_1, \dots, c_n] \quad c_i \in \mathbb{F}^m$$

Let $v = \begin{pmatrix} a_1 \\ \dots \\ a_n \end{pmatrix}$ Then the vector $Av \in \mathbb{F}^m$ is $Av = a_1 c_1 + \dots + a_n c_n$. A different perspective:
if $v = (a_1, \dots, a_n)$, $w = (b_1, \dots, b_n) \in \mathbb{F}^n$. The dot product $v \cdot w = a_1 b_1 + \dots + a_n b_n$. If

$$A = \begin{bmatrix} r_1 \\ \dots \\ r_m \end{bmatrix} \quad r_i \in \mathbb{F}$$

Then $(Av)_i = r_i \cdot v$.

Example 6.1.5

$$A = \begin{pmatrix} 4 & 2 \\ 1 & 0 \end{pmatrix}, v = \begin{pmatrix} 2 \\ -3 \end{pmatrix} \quad Av = 2 \begin{pmatrix} 4 \\ 1 \end{pmatrix} + (-3) \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

Definition 6.1.10

Let $A \in M_{m \times n}(\mathbb{F})$, the function $L_A : \mathbb{F}^n \rightarrow \mathbb{F}^m$ given by $L_A(v) = Av$ is a linear transformation

Proof: Let $v, w \in \mathbb{F}^n$. Let $v = \begin{pmatrix} v_1 \\ \dots \\ v_n \end{pmatrix}$ $w = \begin{pmatrix} w_1 \\ \dots \\ w_n \end{pmatrix}$

$$A(v + w) = A \begin{pmatrix} v_1 + w_1 \\ \dots \\ v_n + w_n \end{pmatrix}$$

if $A = [c_1, \dots, c_n]$

$$\begin{aligned} &= (v_1 + w_1)c_1 + \dots + (v_n + w_n)c_n \\ &= v_1c_1 + \dots + v_nc_n + w_1c_1 + \dots + w_nc_n \\ &= Av + Aw \end{aligned}$$

scalar multiplication left as exercise. ☺

there are also other properties to be checked:

- $L_A = L_B$ iff $A = B$
- If $n = m$ then $L_{I_n} = id$, where I_n is a matrix with 1 on the diagonal top left to bottom right.

Theorem 6.1.10

Let V be an n dimensional vector space, and $\beta = \{v_1, \dots, v_n\}$ be an ordered basis. Then the function $[\cdot]_\beta : V \rightarrow \mathbb{F}^n$ is a linear transformation and is bijective.

Proof: This proof is in HW 7.

Well-definedness and uniqueness of coordinates. Since β is a basis, every $v \in V$ can be written uniquely as

$$v = a_1v_1 + \dots + a_nv_n \quad (a_i \in \mathbb{F}).$$

We define $[v]_\beta := (a_1, \dots, a_n)^\top \in \mathbb{F}^n$.

Linearity. Take $u, v \in V$ and $c \in \mathbb{F}$. Write

$$u = \sum_{i=1}^n \alpha_i v_i, \quad v = \sum_{i=1}^n \beta_i v_i.$$

Then

$$u + v = \sum_{i=1}^n (\alpha_i + \beta_i) v_i \quad \Rightarrow \quad [u + v]_\beta = \begin{pmatrix} \alpha_1 + \beta_1 \\ \vdots \\ \alpha_n + \beta_n \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} + \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix} = [u]_\beta + [v]_\beta.$$

Similarly,

$$cu = \sum_{i=1}^n (c\alpha_i)v_i \Rightarrow [cu]_\beta = \begin{pmatrix} c\alpha_1 \\ \vdots \\ c\alpha_n \end{pmatrix} = c \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = c[u]_\beta.$$

Hence $[\cdot]_\beta$ is linear.

Injectivity. If $[v]_\beta = 0 \in \mathbb{F}^n$, then in the expansion $v = \sum_{i=1}^n a_i v_i$ all coordinates $a_i = 0$ and $v = 0$. For $[v_1]_\beta = [v_2]_\beta$, by linearity, $[v_1]_\beta - [v_2]_\beta = 0 \Rightarrow v_1 - v_2 = 0 \Rightarrow v_1 = v_2$. Therefore map is injective.

Surjectivity. Given any $x = (x_1, \dots, x_n)^\top \in \mathbb{F}^n$, set

$$v := \sum_{i=1}^n x_i v_i \in V.$$

Then, by construction, $[v]_\beta = x$. Thus every vector of $\mathbb{F}^n$ is hit, so the map is surjective.

Since $[\cdot]_\beta$ is linear, injective, and surjective, it is a linear bijection (an isomorphism). ☺

Lemma 6.1.1

Let V, W be finite-dimensional and $\beta = \{v_1, \dots, v_n\}$, $\gamma = \{w_1, \dots, w_m\}$ be ordered bases of V, W . Let $T \in L(V, W)$. Then $[T]_\beta^\gamma$ has $[T(v_j)]_\gamma$ as its j th column.

Proof: By definition, $([T]_\beta^\gamma)_{ij}$ is the scalar a_j such that $T(v_i) = \sum_{j=1}^m a_{ij} w_j$ where $\gamma = \{w_1, \dots, w_m\}$. The j th column of a matrix A are

$$\begin{pmatrix} A_{1j} \\ \vdots \\ A_{mj} \end{pmatrix}$$

It follows that the j th column of $[T]_\beta^\gamma$ is

$$\begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix} = [T(v_j)]_\gamma$$

☺

Theorem 6.1.11 Coordinate rule

Let $T : V \rightarrow W$ be linear, and let $\beta = \{v_1, \dots, v_n\}$ and $\gamma = \{w_1, \dots, w_m\}$ be ordered bases of V and W , respectively. Then for every $v \in V$,

$$[T]_\beta^\gamma [v]_\beta = [T(v)]_\gamma.$$

Proof: Since β is a basis of V , every $v \in V$ can be written uniquely as

$$v = a_1 v_1 + a_2 v_2 + \dots + a_n v_n, \quad a_i \in \mathbb{F}.$$

Hence the coordinate vector of v relative to β is

$$[v]_{\beta} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}.$$

By definition, the matrix of T with respect to β and γ is

$$[T]_{\beta}^{\gamma} = \begin{bmatrix} [T(v_1)]_{\gamma} & [T(v_2)]_{\gamma} & \cdots & [T(v_n)]_{\gamma} \end{bmatrix},$$

that is, each column j of the matrix $[T]_{\beta}^{\gamma}$ is the coordinate vector of $T(v_j)$ in the basis γ .

Because T is linear,

$$T(v) = T(a_1v_1 + \cdots + a_nv_n) = a_1T(v_1) + \cdots + a_nT(v_n).$$

Now we express $T(v)$ in γ -coordinates. Since coordinate mappings are linear,

$$[T(v)]_{\gamma} = [a_1T(v_1) + \cdots + a_nT(v_n)]_{\gamma} = a_1[T(v_1)]_{\gamma} + \cdots + a_n[T(v_n)]_{\gamma}.$$

Notice that this sum exactly matches the definition of matrix multiplication:

$$[T]_{\beta}^{\gamma} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = a_1[T(v_1)]_{\gamma} + \cdots + a_n[T(v_n)]_{\gamma}.$$

Therefore,

$$[T]_{\beta}^{\gamma}[v]_{\beta} = [T(v)]_{\gamma}.$$

This shows that multiplying the coordinate vector of v by the matrix $[T]_{\beta}^{\gamma}$ produces the coordinate vector of $T(v)$ in W . ☺

6.2 Matrix multiplication

Definition 6.2.1

Let $A \in M_{m \times m}(\mathbb{F})$, $B \in M_{m \times p}(\mathbb{F})$. For example,

$$A = \begin{pmatrix} 2 & 3 & 4 \\ 1 & 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 6 \\ -1 & -1 \\ 4 & 0 \end{pmatrix}$$

$$AB = \left(2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ 0 \end{pmatrix} + 4 \begin{pmatrix} 4 \\ 1 \end{pmatrix} \quad 6 \begin{pmatrix} 2 \\ 1 \end{pmatrix} - 1 \begin{pmatrix} 3 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 4 \\ 0 \end{pmatrix} \right)$$

Note:-

If $A \in M_{m \times n}(\mathbb{F})$, $B \in M_{n \times p}(\mathbb{F})$, then $AB \in M_{m \times p}(\mathbb{F})$. Properties:

$$A(B + C) = AB + AC$$

$$(D + E)A = DA + EA$$

$$a(AB) = (aA)B = A(aB) \quad a \in \mathbb{F}$$

$$I_m A = A = A I_n$$

$$D(AC) = (DA)C$$

Does $AB = BA$? Not always. For example

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

$$AB = \begin{pmatrix} 2+0 & 0+0 \\ 0+3 & 0+3 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 3 & 3 \end{pmatrix}$$

$$BA = \begin{pmatrix} 2 & 2 \\ 0 & 3 \end{pmatrix}$$

Theorem 6.2.1

Let V, W, Z be finite-dimensional vector spaces with ordered bases β, γ, δ respectively. Let $T \in L(V, W)$ and $U \in L(W, Z)$. Then

$$[UT]_{\beta}^{\delta} = [U]_{\gamma}^{\delta} [T]_{\beta}^{\gamma}.$$

Proof: Let $\beta = \{v_1, \dots, v_n\}$, $\gamma = \{w_1, \dots, w_m\}$, and $\delta = \{z_1, \dots, z_p\}$. By definition of the matrix of a linear transformation,

$$[T]_{\beta}^{\gamma} = \begin{bmatrix} [T(v_1)]_{\gamma} & [T(v_2)]_{\gamma} & \cdots & [T(v_n)]_{\gamma} \end{bmatrix},$$

and

$$[U]_{\gamma}^{\delta} = \begin{bmatrix} [U(w_1)]_{\delta} & [U(w_2)]_{\delta} & \cdots & [U(w_m)]_{\delta} \end{bmatrix}.$$

We now compute the product $[U]_{\gamma}^{\delta} [T]_{\beta}^{\gamma}$. Matrix multiplication acts column by column, so

$$[U]_{\gamma}^{\delta} [T]_{\beta}^{\gamma} = \begin{bmatrix} [U]_{\gamma}^{\delta} [T(v_1)]_{\gamma} & [U]_{\gamma}^{\delta} [T(v_2)]_{\gamma} & \cdots & [U]_{\gamma}^{\delta} [T(v_n)]_{\gamma} \end{bmatrix}.$$

By Theorem 6.1.11 (the coordinate rule), for any $w \in W$ we have

$$[U]_{\gamma}^{\delta}[w]_{\gamma} = [U(w)]_{\delta}.$$

Applying this to each $T(v_j)$ gives

$$[U]_{\gamma}^{\delta}[T(v_j)]_{\gamma} = [U(T(v_j))]_{\delta} = [(UT)(v_j)]_{\delta}.$$

Substituting this into our matrix product expression,

$$[U]_{\gamma}^{\delta}[T]_{\beta}^{\gamma} = \begin{bmatrix} [(UT)(v_1)]_{\delta} & [(UT)(v_2)]_{\delta} & \cdots & [(UT)(v_n)]_{\delta} \end{bmatrix}.$$

But by definition of the matrix of the composition UT , this last matrix is exactly

$$[UT]_{\beta}^{\delta} = \begin{bmatrix} [(UT)(v_1)]_{\delta} & [(UT)(v_2)]_{\delta} & \cdots & [(UT)(v_n)]_{\delta} \end{bmatrix}.$$

Therefore,

$$[UT]_{\beta}^{\delta} = [U]_{\gamma}^{\delta}[T]_{\beta}^{\gamma}.$$

☺

Definition 6.2.2: Left multiplication transformation

Let A be an $m \times n$ matrix with entries from a field F . We denote by L_A the mapping $L_A : \mathbb{F}^n \rightarrow \mathbb{F}^m$ defined by $L_A(x) = Ax$, which is the matrix product of A and x , for each column vector $x \in \mathbb{F}^n$. We call this **left-multiplication transformation**

Chapter 7

Week 7

Theorem 7.0.1

Let V and W be vector spaces over F with $\dim(V) = n, \dim(W) = m$. The linear transformation $\square_\beta^\gamma : L(V, W) \rightarrow M_{m \times n}(F)$ is bijective.

7.1 Invertibility and isomorphism

Definition 7.1.1: Inverse of linear transformation

Recall the notion of inverse of a function. $T : V \rightarrow W$ is invertible if there exists $S : W \rightarrow V$ such that

$$S \circ T = id_V \quad T \circ S = id_W$$

This inverse is unique and we denote it T^{-1} . Here are some properties

- $(T \circ U)^{-1} = U^{-1} \circ T^{-1}$
- $(T^{-1})^{-1} = T$

Corollary 7.1.1

Let $T : V \rightarrow W$ be a linear transformation with $\dim(V) = \dim(W) < \infty$. Then T is invertible if

$$\text{rank}(T) = \dim(V)$$

iff $\text{null}(T) = 0$.

Example 7.1.1

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$T \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a + b \\ 2a \end{pmatrix}$$

then

$$T^{-1} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{y}{2} \\ x - \frac{y}{2} \end{pmatrix}$$

Theorem 7.1.1 Inverse of linear transformation is linear

Let $T : V \rightarrow W$ be linear and invertible, then T^{-1} is also linear.

Proof: Let's check that T^{-1} preserves scalar multiplication. (It's an exercise). Let $w \in W, c \in \mathbb{F}$. Since T is invertible, there exists a unique $v \in V$ such that $T(v) = w$ and $T^{-1}(w) = v$.

$$T^{-1}(cw) = T^{-1}(cT(v)) = T^{-1}(T(cv)) = cv = cT^{-1}(w)$$

☺

Lemma 7.1.1

Let $T : V \rightarrow W$ be linear and invertible. Then V is finite-dimensional iff W is finite-dimensional. In that case, $\dim(V) = \dim(W)$.

Proof: • $(\Rightarrow)$ Assume $\dim(V) < \infty$. Let β be a basis of V with $|\beta| = n$. Then by surjectivity, $T(\beta) = \{T(v) : v \in \beta\}$ spans W . Since W has a finite spanning set, it has a finite basis and $\dim(W) < \infty$

- $(\Leftarrow)$ For $\dim(W) < \infty \Rightarrow \dim(V) < \infty$, apply the same argument to T^{-1} .
- Now assume $\dim(V) < \infty, \dim(W) < \infty$. By Rank-Nullity theorem,

$$\dim(V) = \text{rank}(T) + \text{null}(T)$$

Since T is injective, $\ker(T) = \{0\} \Rightarrow \text{null}(T) = 0$. Thus

$$\dim(V) = \text{rank}(T) + 0 = \text{rank}(T)$$

Since T is **surjective**, $\text{im}(T) = W$. Thus,

$$\dim(V) = \text{rank}(T) = \dim(\text{im}(T)) = \dim(W)$$

☺

Theorem 7.1.2

Let V and W be finite-dimensional vector spaces with ordered bases β and γ , respectively. Let $T : V \rightarrow W$ be *linear*. Then T is invertible iff $[T]_{\beta}^{\gamma}$ is invertible. Furthermore, $[T^{-1}]_{\gamma}^{\beta} = ([T]_{\beta}^{\gamma})^{-1}$

Proof: $(\Rightarrow)$ Suppose T is invertible, by lemma $\dim(V) = \dim(W)$. Let $n = \dim(V)$, so $[T]_{\beta}^{\gamma}$ is an $n \times n$ matrix. Now $T^{-1} : W \rightarrow V$ satisfies $TT^{-1} = I_W$ and $T^{-1}T = I_V$. Thus

$$I_n = [I_V]_{\beta} = [T^{-1}T]_{\beta} = [T^{-1}]_{\beta}^{\gamma} [T]_{\beta}^{\gamma}$$

Similarly, $[T]_{\beta}^{\gamma} [T^{-1}]_{\gamma}^{\beta} = I_n$. So $[T]_{\beta}^{\gamma}$ is invertible and $([T]_{\beta}^{\gamma})^{-1} = [T^{-1}]_{\gamma}^{\beta}$.

$(\Leftarrow)$ Suppose that $A = [T]_{\beta}^{\gamma}$ is invertible. Then there exists an $n \times n$ matrix B such that $AB = BA = I_n$. Now we want to construct a $[U]_{\gamma}^{\beta} = B$, such that $[U]_{\gamma}^{\beta} [T]_{\beta}^{\gamma} = AB = I_n = [UT]_{\beta}$. Therefore, by decomposing $[U]_{\gamma}^{\beta}$,

$$[U]_{\gamma}^{\beta} = [[U(w_1)]_{\beta}, [U(w_2)]_{\beta}, \dots, [U(w_n)]_{\beta}]$$

For each $[U(w_j)]_\beta$, it's a vector of scalars such that

$$a_{1j}v_1 + a_{2j}v_2 + \dots + a_{jn}v_n = U(w_j)$$

Recalling previous theorem, where it states there exists only one linear transformation such that $T(v_i) = w_i$ where $\beta = \{v_1, v_2, \dots, v_n\}$, $\gamma = \{w_1, w_2, \dots, w_n\}$. So there exists $U \in L(W, V)$ such that Since $B = [U]_\gamma^\beta$, so

$$U(w_j) = B_{1j}v_1 + B_{2j}v_2 + \dots + B_{nj}v_n$$

and

$$U(w_j) = \sum_{i=1}^n B_{ij}v_i$$

To show that $U = T^{-1}$, observe that

$$[UT]_\beta = [U]_\gamma^\beta [T]_\beta^\gamma = BA = I_n = [I_V]_\beta$$

So $UT = I_V$, and similarly, $TU = I_W$. Therefore $U = T^{-1}$.



Definition 7.1.2: Inverse of matrix

Let A be $n \times n$ matrix. Then A is **invertible** if there exists an $n \times n$ matrix B such that $AB = BA = I$. If A is invertible, then $B = A^{-1}$ is **unique**.

Definition 7.1.3: isomorphism

Let V, W be vector spaces. We say that V is **isomorphic** to W if there exists an invertible linear transformation notation $T : V \rightarrow W$. We denote this

$$V \cong W$$

We call an invertible linear transformation an **isomorphism**

Theorem 7.1.3

Let V, W be finite-dimensional vector spaces over $\mathbb{F}$. Then $V \cong W$ iff $\dim(V) = \dim(W)$.

Proof: • $(\Rightarrow)$ Proved in the previous theorem

- $(\Leftarrow)$ Assume that $\dim(V) = \dim(W) = n$. Let $\beta = \{v_1, \dots, v_n\}$ and $\gamma = \{w_1, \dots, w_n\}$ be bases of V, W respectively. Then, by previous theorem, there is a linear transformation T such that $T(v_i) = w_i$. Then $\text{rank}(T) = \dim(\text{span}(T(\beta))) = \dim(\text{span}(\gamma)) = \dim(W)$. So T is surjective. Since $\dim(V) = \dim(W) = \text{rank}(T)$, $\text{null}(T) = 0$ by Rank-Nullity.



Corollary 7.1.2

Let V be a finite-dimensional vector space. Then $V \cong \mathbb{F}^n$, iff $\dim(V) = n$.

Example 7.1.2

Let $\dim(V) = n$ and β be a basis of V . Then

$$[\cdot]_{\beta} : V \rightarrow \mathbb{F}^n$$

is an isomorphism

Proof: TBD



Example 7.1.3

Consider $xy = \{(x, y, z) \in \mathbb{R}^3 | z = 0\}$. It's **not** $\mathbb{R}^2$, but it is isomorphic to $\mathbb{R}^2$.

Theorem 7.1.4

Let V and W be finite-dimensional vector spaces over F of dimensions n and m , respectively, and let β and γ be ordered bases for V and W , respectively. Then the function $[\cdot]_{\beta}^{\gamma} : L(V, W) \rightarrow M_{m \times n}(F)$, is an *isomorphism*

By theorem 6.1.1, $[\cdot]_{\beta}^{\gamma}$ is linear. Hence we only need to show that it's bijective. This is done if we show that for every $m \times n$ matrix A , there exists a unique linear transformation $T : V \rightarrow W$ such that $[T]_{\beta}^{\gamma} = A$. Let $\beta = \{v_1, v_2, \dots, v_n\}$, $\gamma = \{w_1, w_2, \dots, w_m\}$, and let A be a $m \times n$ matrix. By theorem 6.1.1, there exists a unique linear transformation $T : V \rightarrow W$ such that

$$T(v_j) = \sum_{i=1}^m A_{ij} w_i, \quad 1 \leq j \leq n$$

But this means that $[T]_{\beta}^{\gamma} = A$. Thus $[\cdot]_{\beta}^{\gamma}$ is an isomorphism.

An alternative proof (during lecture):

Theorem 7.1.5

Let $\dim(V) = n, \dim(W) = m$, with ordered bases β, γ respectively. Then

$$[\cdot]_{\beta}^{\gamma} : L(V, W) \rightarrow M_{m \times n}(\mathbb{F})$$

Proof: We've shown that $[\cdot]_{\beta}^{\gamma}$ is linear, so it suffices to show that $\ker([\cdot]_{\beta}^{\gamma}) = 0$ and it is surjective.

$$\ker([\cdot]_{\beta}^{\gamma}) = 0$$

Let T be such that $[T]_{\beta}^{\gamma} = 0$. In particular, all elements of $[T]_{\beta}^{\gamma}$ are 0, so

$$T(v_i) = 0w_1 + \dots + 0w_m = 0$$

Since the image of a basis is 0, the image of any vector is 0, and $T = 0 \in L(V, W)$. So $\ker([\cdot]_{\beta}^{\gamma})$

For surjectivity, let $A \in M_{m \times n}(\mathbb{F})$. Then

$$A = (c_1, \dots, c_n)$$

The linear transformation T such that $[T]_\beta^\gamma = A$ is the linear transformation such that

$$T(v_1) = c_{11}w_1 + \dots + c_{m1}w_m$$

By contradiction, $[T]_\beta^\gamma = A$. ☺

Definition 7.1.4: Matrices and inverses

Let $A \in M_{n \times n}(\mathbb{F})$. Then A is invertible iff $\exists B$ s.t. $AB = BA = I_n$. In that case, B is unique and we say $B = A^{-1}$.

Example 7.1.4

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

Theorem 7.1.6

Let $T \in L(V, W)$ with $\dim(V) = \dim(W) = n$. Then T is invertible $\Leftrightarrow [T]_\beta^\gamma$ is invertible (for any choice of ordered bases) and $([T]_\beta^\gamma)^{-1} = [T^{-1}]_\beta^\gamma$.

Proof: T is invertible iff $\exists T^{-1}$ such that $T \circ T^{-1} = id_W$, $T^{-1} \circ T = id_V$.

iff $\exists [T^{-1}]_\beta^\gamma$ s.t.

$$[T]_\beta^\gamma [T^{-1}]_\beta^\gamma = [id_W]_\beta^\gamma = I_n$$

$$[T^{-1}]_\beta^\gamma [T]_\beta^\gamma = [id_V]_\beta^\beta = I_n$$

because $\llbracket \cdot \rrbracket_\beta^\gamma$ is bijective as $[s][v] = [sv]$ ☺

Lemma 7.1.2

Let V, W be finite-dimensional with β, γ as bases. Let $T \in L(V, W)$, let $v \in V$. Then

$$L_{[T]_\beta^\gamma}[v]_\beta = [T(v)]_\gamma$$

Proof:

$$L_{[T]_\beta^\gamma}[v]_\beta = [T]_\beta^\gamma[v]_\beta = [T(v)]_\gamma$$

by the previous theorem. We can think of this as $L_{[T]_\beta^\gamma} \circ \llbracket \cdot \rrbracket_\beta = \llbracket \cdot \rrbracket_\gamma \circ T$

We have $V \rightarrow W$ by T , $\mathbb{F}^n \rightarrow \mathbb{F}^m$ by $L_{[T]_\beta^\gamma}$. $V \rightarrow \mathbb{F}^n$ by $\llbracket \cdot \rrbracket_\beta$, $W \rightarrow \mathbb{F}^m$ by $\llbracket \cdot \rrbracket_\gamma$. We say this relationship is commutative. ☺

Chapter 8

Week 8

8.1 Change of basis

Definition 8.1.1: Change of basis

Let V be finite-dimensional and β, β' be two ordered bases. Then the change of basis matrix from β to β' is $Q = [id]_{\beta'}^{\beta}$

Example 8.1.1

if $V = \mathbb{F}^n, \beta' = st$ then the i th column of $[id]_{\beta'}^{st}$ is the i th vector of β in the standard basis. For example

$$\beta = \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \end{pmatrix} \right\}$$

then

$$[id]_{\beta'}^{st} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$$

Proof: • The identity linear transformation is invertible, so $[id]_{\beta'}^{\beta}$ is an invertible matrix.

- For $v \in V$,

$$Q[v]_{\beta} = [id]_{\beta'}^{\beta}[v]_{\beta} = [id(v)]_{\beta'} = [v]_{\beta'}$$



Theorem 8.1.1

Let β and β' be two ordered bases for a finite-dimensional vector space V , and let $Q = [I_V]_{\beta'}^{\beta}$. Then

- Q is invertible
- For any $v \in V$, $[v]_{\beta} = Q[v]_{\beta'}$

Proof: • Since I_V is invertible, Q is invertible by thm 7.1

- For any $v \in V$,

$$[v]_{\beta} = [I_V(v)]_{\beta} = [I_V]_{\beta'}^{\beta} [v]_{\beta'} = Q[v]_{\beta'}$$



Definition 8.1.2: Change of coordinate matrix

This $Q = [I_V]_{\beta'}^{\beta}$ defined above is called a **change of coordinate matrix**, due to the second part of theorem, we say that Q changes β' coordinates into β coordinates. Observe that if $\beta = \{x_1, x_2, \dots, x_n\}$ and $\beta' = \{x'_1, x'_2, \dots, x'_n\}$, then

$$x'_j = \sum_{i=1}^n Q_{ij} x_i$$

for $j = 1, 2, \dots, n$, that is, the j th column of Q is $[x'_j]_{\beta}$

Theorem 8.1.2

Let $T : V \rightarrow V$ be a linear transformation and β, β' be ordered bases. Let $Q = [id]_{\beta'}^{\beta}$ be the change of basis matrix from β' to β . Then

$$[T]_{\beta'}^{\beta'} = Q^{-1} [T]_{\beta}^{\beta} Q$$

Proof:

$$\begin{aligned} Q[T]_{\beta'}^{\beta'} &= [id]_{\beta'}^{\beta} [T]_{\beta'}^{\beta'} \\ &= [T]_{\beta}^{\beta} \\ &= [T]_{\beta}^{\beta} [id]_{\beta'}^{\beta} = [T]_{\beta}^{\beta} Q \end{aligned}$$

multiply on the left by Q^{-1}



Definition 8.1.3: Linear operator

A **linear operator** is a linear transformation that maps a vector space V into itself.

Corollary 8.1.1

Let $A \in M_{n \times n}(\mathbb{F})$, and let γ be an ordered basis for $\mathbb{F}^n$. Then $[L_A]_{\gamma} = Q^{-1} A Q$, where Q is the $n \times n$ matrix whose j th column is the j th vector of γ

Example 8.1.2

Let

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 3 \\ 0 & -1 & 0 \end{pmatrix},$$

and let

$$\gamma = \left\{ \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\},$$

which is an ordered basis for $\mathbb{R}^3$. Let Q be the 3×3 matrix whose j th column is the j th vector of γ . Then

$$Q = \begin{pmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad Q^{-1} = \begin{pmatrix} -1 & 2 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

So by the preceding corollary,

$$[L_A]_\gamma = Q^{-1}AQ = \begin{pmatrix} 0 & 2 & 8 \\ -1 & 4 & 6 \\ 0 & -1 & -1 \end{pmatrix}.$$

Definition 8.1.4: Similar

We say that two matrices $A, B \in M_{n \times n}(\mathbb{F})$ are similar if there exists an invertible matrix Q s.t.

$$B = Q^{-1}AQ$$

Note:-

This is an equivalence relation

Definition 8.1.5: Diagonalizable

A matrix that is similar to a diagonal matrix is **diagonalizable**. A linear transformation $T : V \rightarrow V$ such that there exists a basis β for which $[T]_\beta$ is diagonal is **diagonalizable**. For example, reflection across a line in $\mathbb{R}^2$.

Definition 8.1.6: Eigenvector and Eigenvalue

Let $T : V \rightarrow V$ be linear, $v \in V, \lambda \in \mathbb{F}$, s.t. $T(v) = \lambda v$ and $v \neq 0$. Then we say that v is an eigenvector of T and λ is the corresponding eigenvalue.

Example 8.1.3

Observation: if $A \in M_{n \times n}(\mathbb{F})$ is a diagonal matrix, then the standard basis vectors are eigenvectors of L_A with eigenvalues $\lambda_1, \dots, \lambda_n$ where $\lambda_1, \dots, \lambda_n$ are elements on the diagonal of A .

Theorem 8.1.3

Let V be finite-dimensional and $T : V \rightarrow V$ be linear. Then T is diagonalizable iff there exists a basis $\beta = \{v_1, \dots, v_n\}$ in which every v_i is an eigenvector.

Proof: ($\Rightarrow$) Assume that T is diagonalizable, then $\exists \beta = \{v_1, \dots, v_n\}$, such that $[T]_\beta$ is a matrix with diagonal elements $\lambda_1, \dots, \lambda_n$. But $[T]_\beta = ([T(v_1)]_\beta \dots [T(v_n)]_\beta)$ so $T(v_i) = 0v_1 + \dots + \lambda_i v_i + \dots + 0v_n = \lambda_i v_i$. So v_i is an eigenvector with eigenvalue λ_i .

($\Leftarrow$) Assume that $\exists$ a basis of eigenvectors, $\beta = \{v_1, \dots, v_n\}$. Then

$$T(v_i) = \lambda_i v_i = \lambda_i v_i + \sum_{j \neq i} 0v_j$$

so

$$[T(v_i)]_\beta = \begin{pmatrix} 0 \\ \vdots \\ \lambda_i \\ \vdots \\ 0 \end{pmatrix}$$

and $[T]_\beta$ is a diagonal matrix with $\lambda_1, \dots, \lambda_n$ on diagonal.



Chapter 9

Week 9

9.1 Matrix inversion

Definition 9.1.1

An elementary matrix is square matrix that differs from I_n in one of three ways:

1. Two rows are swapped, e.g.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

2. One diagonal entry has linear multiplied by a nonzero scalar

$$\begin{pmatrix} 7 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

3. One off-diagonal entry has been given a nonzero value, e.g.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 11 \\ 0 & 0 & 1 \end{pmatrix}$$

Example 9.1.1

Let $A \in M_{n \times n}(\mathbb{F})$ and let $E \in M_{n \times n}(\mathbb{F})$ be an elementary matrix. Describe how to obtain the matrix EA from the matrix A if E is type 1, 2 and 3?

1. E is type 1 with rows i and j supposed then EA is obtained from A by swapping rows i and j
2. E is type 2 with $E_{ii} = \lambda$, then the EA is obtained from A by multiplying the i th row of A by λ
3. If E is of type 3 with $\lambda = E_{ij} \neq 0$, $i \neq j$, then EA is obtained from A by adding $\lambda(j$ th row of A) to i th row of A

Definition 9.1.2

The elementary row operation:

1. swapping two rows
2. multiplying the row by a nonzero scalar
3. adding a nonzero multiple of one row to another row

Proposition 9.1.1

If E is of type 1,2,3, then EA is obtained from A by the row operation 1,2,3.

Proof:



Corollary 9.1.1

If E is elementary, then E is invertible, and E^{-1} is the same type to E

Proof: 1. Type 1, $EE = I_n$ because multiplication by E on the left "unswapes" the two rows.

2. Type 2, E^{-1} has the same entry i, i in E modified, with value $E_{ii}^{-1} = (E_{ii})^{-1}$

3. Type 3, if $E_{ij} \neq 0, i \neq j$, then $E_{ij}^{-1} = -(E_{ij})$



Definition 9.1.3

$$\text{rank}(A) = \dim(L_A)$$

Proposition 9.1.2

$$\text{rank}(A) = \dim(\text{span}(\text{columns of } A))$$

Proof:



Corollary 9.1.2

The columns of an invertible matrix $A \in M_{n \times n}(\mathbb{F})$ form a basis of $\mathbb{F}^n$. In particular, they are linearly independent.

Proof: A spanning set of size n for $\mathbb{F}^n$ is a basis.



Theorem 9.1.1

Let V, W, Z be finite-dimensional vector spaces, and $T : V \rightarrow W, U : W \rightarrow Z$ be linear transformations, prove

- $\text{rank}(UT) \leq \text{rank}(T)$

- $\text{rank}(UT) \leq \text{rank}(U)$

Moreover, let A, B be matrices such that the product AB is defined, prove

- $\text{rank}(AB) \leq \text{rank}(A)$
- $\text{rank}(AB) \leq \text{rank}(B)$

Proof: 1.

$$\dim(V) = \text{rank}(UT) + \text{null}(UT) = \text{rank}(T) + \text{null}(T)$$

if $T(v) = 0 \Rightarrow UT(v) = 0 \Rightarrow \ker(T) \subset \ker(UT)$,

$$\text{null}(UT) \geq \text{null}(T) \Rightarrow \text{rank}(UT) \leq \text{rank}(T)$$

2. $\text{im}(T) \subseteq W$

$$U(\text{im}(T)) \subseteq U(W)$$

$$\text{im}(UT) \subseteq \text{im}(U)$$

$$\text{rank}(UT) \leq \text{rank}(U)$$

☺

Theorem 9.1.2

Let $A \in M_{n \times n}(\mathbb{F})$, $P \in M_{m \times m}(\mathbb{F})$, $Q \in M_{n \times n}(\mathbb{F})$ such that P, Q are invertible. Then

$$\text{rank}(A) = \text{rank}(PA) = \text{rank}(PAQ)$$

Proof: By the exercise, $\text{rank}(PA) \leq \text{rank}(A)$. But P is invertible, so P^{-1} exists. $\text{rank}(A) = \text{rank}(P^{-1}PA) \leq \text{rank}(PA)$ (applying previous result to P^{-1})

$$\Rightarrow \text{rank}(A) = \text{rank}(PA)$$

the other equalities are proved the same way.

☺

Theorem 9.1.3

Let $A \in M_{n \times n}(\mathbb{F})$ be invertible. Then there exists elementary matrices $E_1, E_2, \dots, E_r$ such that

$$E_r \dots E_2 E_1 A = I_n$$

or, in other words, $E_r \dots E_2 E_1 = A^{-1}$

Proof: Let A be an invertible $n \times n$ matrix. We proceed column by column to construct invertible elementary matrices that reduce A to the identity.

Step 1. If the top-left entry a_{11} is zero, use a *Type 1* elementary matrix (row swap) to interchange rows so that the entry in the first column becomes nonzero. This is possible because A is invertible, so its first column cannot be the zero vector.

Step 2. Use a *Type 2* elementary matrix (row scaling) to multiply the first row by a scalar so that the top-left entry becomes 1.

Step 3. Use *Type 3* elementary matrices (row addition) to eliminate all entries below the pivot in the first column. After this step, the first column is of the form

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \end{bmatrix}^T.$$

Step 4. Focus on the submatrix obtained by deleting the first row and first column. If the pivot position in this submatrix is zero, use a Type 1 swap within that submatrix to make it nonzero, then repeat Steps 2 and 3. Because A is invertible, the pivot column at each stage cannot be a linear combination of the previous ones, so we can always make a nonzero pivot.

Step 5. Continue this process for each successive column. Since there are finitely many columns, the procedure terminates after finitely many steps, producing the identity matrix. Each operation corresponds to multiplication on the left by an invertible elementary matrix E_i , so

$$E_k E_{k-1} \cdots E_1 A = I.$$

Hence $A^{-1} = E_k E_{k-1} \cdots E_1$, and every step preserves invertibility and rank. ☺

Example 9.1.2 (Example)

Consider

$$A = \begin{bmatrix} 0 & 2 & 7 \\ 2 & 4 & 6 \\ 3 & 0 & 1 \end{bmatrix}.$$

1. The top-left entry is 0, so we swap the first and second rows (Type 1):

$$R_1 \leftrightarrow R_2 \quad \Rightarrow \quad \begin{bmatrix} 2 & 4 & 6 \\ 0 & 2 & 7 \\ 3 & 0 & 1 \end{bmatrix}.$$

2. Scale the first row so that the pivot becomes 1 (Type 2):

$$R_1 \rightarrow \frac{1}{2}R_1 \quad \Rightarrow \quad \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 7 \\ 3 & 0 & 1 \end{bmatrix}.$$

3. Eliminate the entry below the pivot (Type 3):

$$R_3 \rightarrow R_3 - 3R_1 \quad \Rightarrow \quad \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 7 \\ 0 & -6 & -8 \end{bmatrix}.$$

4. Move to the 2×2 submatrix in the lower right and repeat the same steps (swap, scale, eliminate). Continuing the process eventually yields the identity.

Each operation above corresponds to left-multiplying A by an invertible elementary matrix. Therefore,

$$E_k E_{k-1} \cdots E_1 A = I \quad \text{and} \quad A^{-1} = E_k E_{k-1} \cdots E_1.$$

9.2 Systems of Linear Equation

Theorem 9.2.1

Finding a solution $x_1, \dots, x_n$ to this system accounts to finding a vector $x = \begin{pmatrix} x_1 \\ \dots \\ x_n \end{pmatrix} \in \mathbb{F}^n$ such that $Ax = b$, $b = \begin{pmatrix} b_1 \\ \dots \\ b_m \end{pmatrix} \in \mathbb{F}^m$.

Proof:



Proposition 9.2.1

If A is invertible, (in particular $A \in M_{n \times n}(\mathbb{F})$)

1. The system $Ax = b$ has a unique solution
2. The solution is obtained from b by applying the same row operations used to invert A

Proof: 1. If A is invertible, then the vector $A^{-1}b$ satisfies $A(A^{-1}b) = b$ so it is a solution. It is unique because L_A is injective.
2. By previous theorem, $A^{-1} = E_r \dots E_2 E_1$ for E_i elementary, so the solution $A^{-1}b$ is $E_r \dots E_2 E_1 b$, i.e. the same sequence of row operations, applied to b .



Theorem 9.2.2

Let $Ax = b$ be a system of linear equations, with $A \in M_{m \times n}(\mathbb{F})$.

1. The system has a solution for every b iff $\text{rank}(A) = m$ or L_A is surjective.
2. Assume $Ax = b$ has a solution, it is unique iff $\text{null}(A) = 0$ or L_A is injective.

Proof: 1. $Ax = b$ has a solution if $b \in \text{im}(L_A)$. This will hold for every $b \in \mathbb{F}^m$ iff $\text{im}(L_A) = \mathbb{F}^m$, i.e. L_A is surjective, which means $\text{rank}(A) = m$.
2. Assuming $Ax = b$, then by linearity of L_A , $A(x + v) = b$ iff $Av = 0$. So x is unique iff the only v such that $Av = 0$ is $v = 0$, i.e. $\text{null}(A) = 0$.



Definition 9.2.1

The system of the form $Ax = 0$ is called **homogenous**

Theorem 9.2.3

The system $Ax = 0$ has a unique solution iff the columns of A are linearly independent.

Proof: Recall that if $v = \begin{pmatrix} v_1 \\ \dots \\ v_n \end{pmatrix}$ and

$$A = \begin{pmatrix} c_1 \\ \dots \\ c_n \end{pmatrix}$$

where c_i are row vectors. Then

$$Av = v_1c_1 + \dots + v_nc_n$$

so $Av = 0$ for $v \neq 0$ iff $\exists$ a nontrivial relation between the columns. So by the previous result, the columns are linearly independent iff $\ker(A) = 0$ iff $Ax = 0$ has a unique solution. ☺

Question 1

Recall the algorithm to invert an invertible matrix. If A is not invertible (in particular, not necessarily a square) what could go wrong?

1. You arrive at a column that is a linear combination of the columns to its left:

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

2. You've exhausted your columns in which you can put 1's, but you still have rows left:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Definition 9.2.2

A matrix $A \in M_{m \times n}(F)$ is in *reduced row echelon form* if:

1. The first nonzero entry of every row is a 1, or the row is a row of zeros. We call these 1's the *leading 1's*.
2. Every leading 1 has only zeros above and below it, and it is to the right of any leading 1 in a row above it.
3. Every row of zeros is at the bottom.

Theorem 9.2.4

Let $A \in M_{m \times n}(F)$. Then there exist $E_1, \dots, E_r \in M_{m \times m}(F)$, elementary, such that

$$E_r \cdots E_1 A \text{ is in RREF.}$$

Proof: We run the algorithm to “invert A ”. If we arrive at a column which is a linear combination of the previous ones (in particular, its lower entries are all zero), then skip it.

It may also be that you run out of columns. Let us say that you have k leading 1's. Then every column is a linear combination of $e_1, \dots, e_k$, so all the lower entries ($i > k$) are zero in all columns.

So the matrix is in RREF. ☺

Theorem 9.2.5 Rank and Leading Ones

The rank of A is the number of leading 1's of $E_r \cdots E_1 A$.

Proof: We have

$$\text{rank}(E_r \cdots E_1 A) = \text{rank}(A),$$

because each E_i is invertible.

Let

$$P = E_r \cdots E_1.$$

The columns of PA are the union of two sets:

$$C_1 = \{\text{columns with a leading 1}\},$$

$$C_{\text{LD}} = \{\text{columns skipped because they were linearly dependent on earlier ones}\}.$$

By our earlier theorem,

$$\text{span}(C_1 \cup \{c\}) = \text{span}(C_1) \quad \text{for all } c \in C_{\text{LD}}.$$

Hence,

$$\text{span}(C_1) = \text{span}(C_1 \cup C_{\text{LD}}).$$

Once the columns with leading 1's are in RREF, they are standard basis vectors. Thus they are linearly independent and form a basis of $\text{im}(PA)$.

Therefore,

$$\text{rank}(A) = \text{rank}(PA) = |C_1| = \# \text{ of leading 1's}.$$

☺

Chapter 10

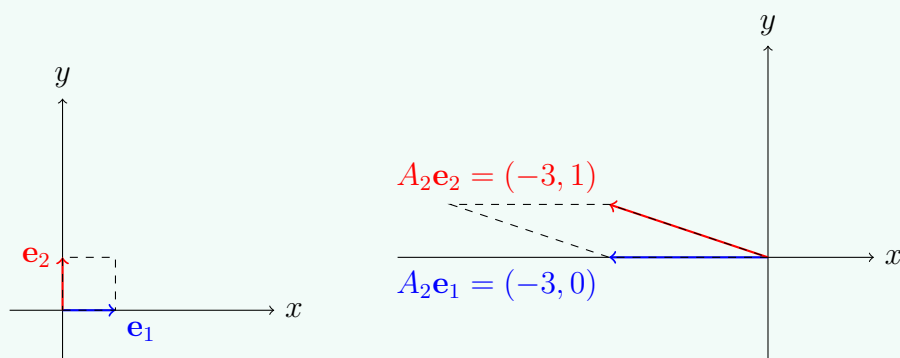
Week 10

10.1 Determinant

Example 10.1.1

Let

$$A_2 = \begin{pmatrix} -3 & -3 \\ 0 & 1 \end{pmatrix}.$$



The area of the image parallelogram is

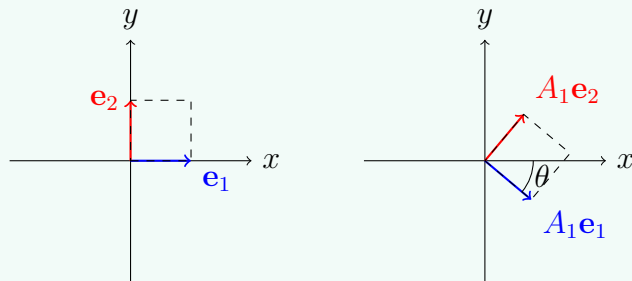
$$3 \times 1 = 3,$$

so $|\det(A_2)| = 3$.

Example 10.1.2

Let

$$A_1 = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$



We compute

$$\det(A_1) = \cos^2 \theta + \sin^2 \theta = 1,$$

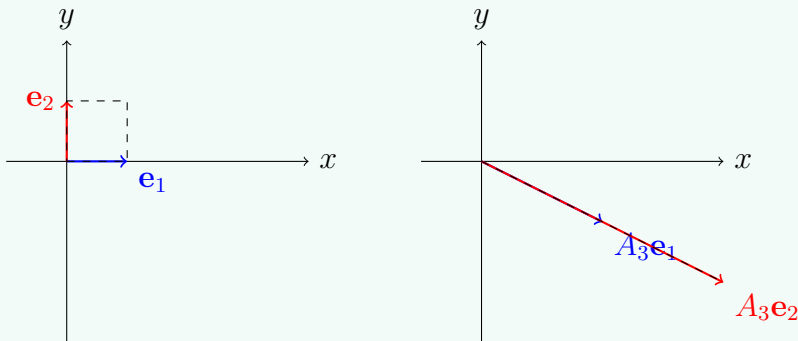
so the area of the unit square is preserved.

Example 10.1.3

Let

$$A_3 = \begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix}.$$

$$\det(A_3) = 2(-2) - 4(-1) = -4 + 4 = 0.$$



Both $A_3\mathbf{e}_1$ and $A_3\mathbf{e}_2$ lie on the same line, so the image of the unit square is a line segment; its area is 0, matching $\det(A_3) = 0$.

Definition 10.1.1: 2x2 determinant

The determinant of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $ad - bc$.

Definition 10.1.2: Minor of a Matrix

Let A be an $n \times n$ matrix over $\mathbb{F}$. The minor of the entry A_{ij} , denoted $\tilde{A}_{ij}$, is the matrix obtained from A by deleting the i th row and the j th column of A .

Example 10.1.4

Let

$$A = \begin{pmatrix} 4 & 1 & 1 & 1 \\ 1 & 0 & 0 & 2 \\ 2 & 0 & 0 & 2 \\ 3 & 6 & 1 & 4 \end{pmatrix}.$$

Then the minor $\tilde{A}_{11}$ is obtained by removing the first row and first column:

$$\tilde{A}_{11} = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 2 \\ 6 & 1 & 4 \end{pmatrix}.$$

Similarly, the minor $\tilde{A}_{24}$ is obtained by removing the second row and fourth column:

$$\tilde{A}_{24} = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 0 & 0 \\ 3 & 6 & 1 \end{pmatrix}.$$

Definition 10.1.3: Laplace expansion

Let $A \in M_n(\mathbb{F})$.

- If $n = 1$, then $\det(A) = A_{11}$.
- If $n > 1$, then

$$\det(A) = \sum_{j=1}^n (-1)^{1+j} A_{1j} \det(\tilde{A}_{1j}).$$

Example 10.1.5

Let

$$A = \begin{pmatrix} 4 & 0 & 1 & 1 \\ 1 & 0 & 0 & 2 \\ 2 & 0 & 0 & 2 \\ 3 & 6 & 1 & 4 \end{pmatrix}.$$

Then

$$\det(A) = 4 \det \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 2 \\ 6 & 1 & 4 \end{pmatrix} - 0 \det \begin{pmatrix} 1 & 0 & 2 \\ 2 & 0 & 2 \\ 3 & 6 & 4 \end{pmatrix} + 1 \det \begin{pmatrix} 1 & 0 & 2 \\ 2 & 0 & 2 \\ 3 & 6 & 4 \end{pmatrix} - 1 \det \begin{pmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 3 & 6 & 1 \end{pmatrix}.$$

Compute each minor:

$$\det \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 2 \\ 6 & 1 & 4 \end{pmatrix} = 0 - 0 + 2 \det \begin{pmatrix} 0 & 0 \\ 6 & 1 \end{pmatrix} = 2(0 - 0) = 0.$$

$$\det \begin{pmatrix} 1 & 0 & 2 \\ 2 & 0 & 2 \\ 3 & 6 & 4 \end{pmatrix} = 1 \det \begin{pmatrix} 0 & 2 \\ 6 & 4 \end{pmatrix} - 0 + 2 \det \begin{pmatrix} 2 & 0 \\ 3 & 6 \end{pmatrix}.$$

The subdeterminants are

$$\det \begin{pmatrix} 0 & 2 \\ 6 & 4 \end{pmatrix} = 0 \cdot 4 - 2 \cdot 6 = -12, \quad \det \begin{pmatrix} 2 & 0 \\ 3 & 6 \end{pmatrix} = 2 \cdot 6 - 0 = 12.$$

Thus

$$\det \begin{pmatrix} 1 & 0 & 2 \\ 2 & 0 & 2 \\ 3 & 6 & 4 \end{pmatrix} = -12 + 2(12) = -12 + 24 = 12.$$

Finally,

$$\det \begin{pmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 3 & 6 & 1 \end{pmatrix} = 1 \det \begin{pmatrix} 0 & 0 \\ 6 & 1 \end{pmatrix} = 0.$$

Putting all terms together:

$$\det(A) = 4(0) - 0 + 1(12) - 1(0) = 12.$$

Example 10.1.6 (Diagonal Matrix)

If $D = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{pmatrix}$ is a diagonal matrix, then

$$\det(D) = \prod_{i=1}^n \lambda_i$$

Question 1

Prove or disprove: Let $M_{n \times n}(F) \rightarrow F$ be a linear transformation.

To disprove, there could be such counter example:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}$$

$$\det(A) = 1 \quad \det(B) = 0 \quad \det(A + B) = 2$$

$$1 + 0 \neq 2$$

Theorem 10.1.1 Linearity in a Row

The determinant is linear in each row separately. If the k th row of an $n \times n$ matrix A is

replaced by $r_k + cr'_k$, then

$$\det \begin{pmatrix} r_1 \\ \vdots \\ r_k + cr'_k \\ \vdots \\ r_n \end{pmatrix} = \det \begin{pmatrix} r_1 \\ \vdots \\ r_k \\ \vdots \\ r_n \end{pmatrix} + c \det \begin{pmatrix} r_1 \\ \vdots \\ r'_k \\ \vdots \\ r_n \end{pmatrix}.$$

Proof: We proceed by induction on n .

Base case $n = 1$. A 1×1 matrix is just (a) , and

$$\det(a + ca') = a + ca' = \det(a) + c \det(a'),$$

so the result holds.

Inductive step. Assume the statement holds for all $(n - 1) \times (n - 1)$ matrices. Let A be an $n \times n$ matrix whose k th row is replaced by $r_k + cr'_k$.

There are two cases.

Case 1: $k = 1$. Expanding along the first row,

$$\det(A_{\text{modified}}) = \sum_{i=1}^n (-1)^{1+i} (r_1 + cr'_1)_i \det(\tilde{A}_{1i}).$$

Distribute inside the sum:

$$\sum_{i=1}^n (-1)^{1+i} r_{1i} \det(\tilde{A}_{1i}) + c \sum_{i=1}^n (-1)^{1+i} r'_{1i} \det(\tilde{A}_{1i}) = \det(A) + c \det(A').$$

Case 2: $k > 1$.

Expand along the first row:

$$\det(A_{\text{modified}}) = \sum_{j=1}^n (-1)^{1+j} A_{1j} \det(\tilde{A}_{1j}).$$

Elaboration: Why the induction hypothesis applies to each minor.

Inside each minor $\tilde{A}_{1j}$, the first row of A has been removed. Thus all rows below row 1 shift upward by one position:

- row 2 of A becomes row 1 of $\tilde{A}_{1j}$ - row 3 of A becomes row 2 of $\tilde{A}_{1j}$ - - row k of A becomes row $k - 1$ of $\tilde{A}_{1j}$.

Since the k th row of A was replaced by $r_k + cr'_k$, the $(k-1)$ st row of each minor is

$$r_k + cr'_k.$$

Therefore each minor has exactly the same structure as the original problem, but it is an $(n-1) \times (n-1)$ matrix with its $(k-1)$ st row modified.

Now assume $k > 1$. Then the minors satisfy the theorem by the induction hypothesis. (A “ $\hat{}$ ” over an index means skipping that entry.)

$$\det \begin{pmatrix} r_1 \\ \vdots \\ r_k + cr'_k \\ \vdots \\ r_n \end{pmatrix} = \sum_{j=1}^n (-1)^{1+j} A_{1j} \det \begin{pmatrix} r_{2\hat{j}} \\ \vdots \\ r_{k\hat{j}} + cr'_{k\hat{j}} \\ \vdots \\ r_{n\hat{j}} \end{pmatrix}.$$

By the induction hypothesis,

$$\det \begin{pmatrix} r_{2\hat{j}} \\ \vdots \\ r_{k\hat{j}} + cr'_{k\hat{j}} \\ \vdots \\ r_{n\hat{j}} \end{pmatrix} = \det \begin{pmatrix} r_{2\hat{j}} \\ \vdots \\ r_{k\hat{j}} \\ \vdots \\ r_{n\hat{j}} \end{pmatrix} + c \det \begin{pmatrix} r_{2\hat{j}} \\ \vdots \\ r'_{k\hat{j}} \\ \vdots \\ r_{n\hat{j}} \end{pmatrix}.$$

Therefore,

$$\begin{aligned} \det \begin{pmatrix} r_1 \\ \vdots \\ r_k + cr'_k \\ \vdots \\ r_n \end{pmatrix} &= \sum_{j=1}^n (-1)^{1+j} A_{1j} \left[\det \begin{pmatrix} r_{2\hat{j}} \\ \vdots \\ r_{k\hat{j}} \\ \vdots \\ r_{n\hat{j}} \end{pmatrix} + c \det \begin{pmatrix} r_{2\hat{j}} \\ \vdots \\ r'_{k\hat{j}} \\ \vdots \\ r_{n\hat{j}} \end{pmatrix} \right] \\ &= \sum_{j=1}^n (-1)^{1+j} A_{1j} \det \begin{pmatrix} r_{2\hat{j}} \\ \vdots \\ r_{k\hat{j}} \\ \vdots \\ r_{n\hat{j}} \end{pmatrix} + c \sum_{j=1}^n (-1)^{1+j} A_{1j} \det \begin{pmatrix} r_{2\hat{j}} \\ \vdots \\ r'_{k\hat{j}} \\ \vdots \\ r_{n\hat{j}} \end{pmatrix}. \end{aligned}$$

The first sum equals $\det(A)$ and the second equals $\det(A')$, where A' is A with row k replaced by r'_k . Hence,

$$\det(A_{\text{modified}}) = \det(A) + c \det(A').$$



Corollary 10.1.1 If A has a row of zeros, then $\det(A) = 0$.

Let

$$A = \begin{pmatrix} r_1 \\ \vdots \\ 0 \\ \vdots \\ r_n \end{pmatrix}.$$

Then, by linearity in the zero row,

$$\det(A) = \det \begin{pmatrix} r_1 \\ \vdots \\ 0 \\ \vdots \\ r_n \end{pmatrix} = \det \begin{pmatrix} r_1 \\ \vdots \\ 0 \\ \vdots \\ r_n \end{pmatrix} - \det \begin{pmatrix} r_1 \\ \vdots \\ 0 \\ \vdots \\ r_n \end{pmatrix} = 0.$$

Example 10.1.7 (Lower triangular matrices)

For example:

$$\begin{pmatrix} 2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 0 \\ 4 & -1 & -1 & 9 \\ 0 & 2 & 11 & 0 \end{pmatrix}.$$

Theorem 10.1.2

Let A' be obtained from A by swapping two adjacent rows, then

$$\det(A') = -\det(A)$$

Proof: Assume

$$A' = \begin{pmatrix} r_2 \\ r_1 \\ \vdots \\ r_n \end{pmatrix}$$

Then

$$\begin{aligned} \det(A') &= \sum_{j=1}^n (-1)^{j+1} A_{2j} \det(\widetilde{A_{2j}}) \\ &= \sum_{j=1}^n (-1)^{j+1} A_{2j} \cdot \left(\sum_{i=1}^{j-1} (-1)^{i+1} A_{1i} \det(\widetilde{(\widetilde{A_{2j}})_{1i}}) \right) + \sum_{i=j+1}^n (-1)^i A_{1i} \det(\widetilde{(\widetilde{A_{2j}})_{ij}}) \\ &= \sum_{j=1}^n (-1)^{j+1} A_{2j} \cdot \left(\sum_{i=1}^{j-1} (-1)^{i+1} A_{1i} \det(\widetilde{((\widetilde{A_{2j}})_{1i})}) \right) + \sum_{i=j+1}^n (-1)^i A_{1i} \det(\widetilde{((\widetilde{A_{2j}})_{1i})}) \end{aligned}$$

Observe that

$$\begin{aligned}
& ((\widetilde{A_{2j}})_{1i})) = ((\widetilde{A_{1i}})_{2j})) \\
&= \sum_{i < j} (-1)^{i+j+2} A_{2j} A_{1i} \det(((\widetilde{A_{1i}})_{2j}))) + \sum_{i > j} (-1)^{i+j+1} A_{2j} A_{1i} \det(((\widetilde{A_{1i}})_{2j}))) \\
&= (-1) \left(\sum_{i < j} (-1)^{i+j+1} A_{2j} A_{1i} \det(((\widetilde{A_{1i}})_{2j}))) + \sum_{i > j} (-1)^{i+j+2} A_{2j} A_{1i} \det(((\widetilde{A_{1i}})_{2j}))) \right) \\
&= -1 \sum_{i=1}^n (-1)^{i+1} A_{1i} \cdot \left(\sum_{j=1}^{i-1} (-1)^{j+1} A_{2j} \det(((\widetilde{A_{1i}})_{2j}))) + \sum_{j=i+1}^n (-1)^i A_{2j} \det(((\widetilde{A_{1i}})_{2j}))) \right) \\
&= -\det(A)
\end{aligned}$$

If the two swapped rows are k and $k+1$, then the minors in the k th iteration will all satisfy

$$\det(M') = -\det(M),$$

by induction. ☺

Corollary 10.1.2

If A has two identical rows, then $\det(A) = 0$.

Proof: Let A' be obtained from A by swapping the two identical rows. Then

$$\det(A') = -\det(A),$$

but $A' = A$, hence

$$\det(A) = -\det(A).$$

Thus $\det(A) = 0$. ☺

Corollary 10.1.3

If one row of A is a scalar multiple of another row, then $\det(A) = 0$.

Proof: Suppose row j equals λ times row i :

$$r_j = \lambda r_i.$$

Replace row j by $r_j - \lambda r_i$. This operation is a Type III row operation (adding a multiple of one row to another), and it does **not** change the determinant:

$$\det(A) = \det(A').$$

But in A' , row j becomes

$$r_j - \lambda r_i = 0,$$

so A' has a zero row, hence

$$\det(A') = 0.$$

Thus $\det(A) = 0$. ☺

Corollary 10.1.4

If E is an elementary matrix of Type I (row swap), then $\det(E) = -1$.

Proof: A Type I elementary matrix performs a **single swap of two adjacent rows**. We already proved that swapping two rows changes the sign of the determinant:

$$\det(EA) = -\det(A).$$

Apply this to $A = I$:

$$\det(E) = \det(EI) = -\det(I).$$

Since $\det(I) = 1$, we have

$$\det(E) = -1.$$
☺

Theorem 10.1.3 Row addition preserves determinant

Let A' be obtained from A by adding a multiple of one row to another row. Then

$$\det(A') = \det(A).$$

Proof: Assume

$$A = \begin{pmatrix} r_1 \\ \vdots \\ r_i \\ \vdots \\ r_n \end{pmatrix}, \quad A' = \begin{pmatrix} r_1 \\ \vdots \\ r_i + cr_j \\ \vdots \\ r_n \end{pmatrix}, \quad i \neq j.$$

Using linearity of the determinant in the i -th row,

$$\det(A') = \det \begin{pmatrix} r_1 \\ \vdots \\ r_i + cr_j \\ \vdots \\ r_n \end{pmatrix} = \det \begin{pmatrix} r_1 \\ \vdots \\ r_i \\ \vdots \\ r_n \end{pmatrix} + c \det \begin{pmatrix} r_1 \\ \vdots \\ r_j \\ \vdots \\ r_n \end{pmatrix}.$$

But in the second determinant, rows i and j are identical, so that determinant is zero. Therefore,

$$\det(A') = \det(A) + c \cdot 0 = \det(A).$$
☺

Corollary 10.1.5

$\det(A)$ can be computed by the cofactor expansion (or Laplace expansion) along any row,

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij}).$$

Proof: Start with the matrix

$$A = \begin{pmatrix} r_1 \\ \vdots \\ r_i \\ \vdots \\ r_n \end{pmatrix}.$$

By swapping row r_i upward $i - 1$ times, we obtain the matrix

$$A' = \begin{pmatrix} r_i \\ r_1 \\ \vdots \\ r_{i-1} \\ \vdots \\ r_n \end{pmatrix}.$$

Observe that

$$\tilde{A}'_{ij} = \tilde{A}_{ij} \quad \text{and} \quad A'_{ij} = A_{ij}.$$

Since each row swap contributes a factor of (-1) , and we perform $i - 1$ swaps,

$$\det(A) = (-1)^{i-1} \det(A').$$

Now apply cofactor expansion along the first row of A' :

$$\det(A') = \sum_{j=1}^n (-1)^{1+j} A'_{1j} \det(\tilde{A}'_{1j}),$$

so

$$\det(A) = (-1)^{i-1} \sum_{j=1}^n (-1)^{1+j} A_{ij} \det(\tilde{A}_{ij}) = \sum_{j=1}^n (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij}).$$

Thus the cofactor expansion formula holds along any row.

□



This method is extremely useful! Always try to expand along a row or column containing many zeros. Do not forget the alternating signs. It is often useful to visualize the sign matrix:

$$\begin{pmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{pmatrix}.$$

Example 10.1.8

Compute the determinant:

$$\det \begin{pmatrix} 2 & 4 & 6 & 1 & 2 \\ 3 & 0 & 6 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 2 & 11 & 4 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix}.$$

Expanding along row 3 (which contains many zeros):

$$= -1 \cdot \det \begin{pmatrix} 2 & 4 & 1 & 2 \\ 3 & 0 & 0 & 0 \\ 0 & 2 & 4 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

Now expand along row 2:

$$= (-1)(-3) \cdot \det \begin{pmatrix} 4 & 1 & 2 \\ 2 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Expand along the last row:

$$= (-1)(-3) \cdot 1 \cdot \det \begin{pmatrix} 4 & 1 \\ 2 & 4 \end{pmatrix}.$$

Finally,

$$(-1)(-3) \cdot 1 \cdot (4 \cdot 4 - 2 \cdot 1) = 3(16 - 2) = 3 \cdot 14 = 42.$$

Theorem 10.1.4

If E is an elementary matrix, then $\det(EA) = \det(E) \det(A)$.

Proof: **Type 1 (row swap).** If E swaps two rows of A , then $\det(E) = -1$. Since row-swapping multiplies the determinant by -1 ,

$$\det(EA) = -\det(A) = \det(E) \det(A).$$

Type 2 (row scaling). If E multiplies row i by a nonzero constant c , then

$$\det(E) = c.$$

The matrix EA is obtained from A by multiplying a row by c , so

$$\det(EA) = c \det(A) = \det(E) \det(A).$$

Type 3 (row replacement). If E is obtained by adding a multiple of one row to another, then

$$\det(E) = 1.$$

Since adding a multiple of one row to another does not change the determinant,

$$\det(EA) = \det(A) = 1 \cdot \det(A) = \det(E) \det(A).$$

Thus, in all three cases,

$$\det(EA) = \det(E) \det(A).$$



Theorem 10.1.5 $\det(A) = 0$ iff A is not invertible

A square matrix is invertible if and only if its determinant is nonzero.

Proof: ($\Rightarrow$) Suppose first that A is invertible. Then there exist elementary matrices $E_1, \dots, E_k$ such that

$$A = E_1 \cdots E_k I.$$

Using multiplicativity of the determinant,

$$\det(A) = \det(E_1) \cdots \det(E_k) \det(I).$$

Each $\det(E_i) \neq 0$, and $\det(I) = 1$, hence $\det(A) \neq 0$.

($\Leftarrow$): Now suppose A is not invertible. Then the columns of A are linearly dependent, so

$$\text{rank}(A) < n.$$

The rank of a matrix equals the number of pivot rows in its RREF. Thus, $\text{RREF}(A)$ has fewer than n pivot rows. Since an $n \times n$ matrix has n rows total, having fewer than n pivot rows means that at least one row contains no pivot. In RREF, every non-pivot row is necessarily a zero row. Therefore,

$$\det(\text{RREF}(A)) = 0.$$

Which can be deduced by cofactor expansion on the zero row.

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} \underbrace{A_{ij}}_{=0} \det(\tilde{A}_{ij}).$$

Because $\text{RREF}(A)$ is obtained from A by a product of elementary matrices,

$$\text{RREF}(A) = E_1 \cdots E_\ell A,$$

so

$$A = E_\ell^{-1} \cdots E_1^{-1} \text{RREF}(A).$$

Taking determinants gives

$$\det(A) = \det(E_\ell^{-1}) \cdots \det(E_1^{-1}) \det(\text{RREF}(A)) = 0.$$

Thus $\det(A) = 0$ exactly when A is not invertible. ☺

Theorem 10.1.6 $\det(AB) = \det(A) \det(B)$

Proof: If A is not invertible, then $\det(A) \det(B) = 0$. Also, $\text{rank}(AB) \leq \text{rank}(A)$, so AB is also not invertible. Therefore,

$$\det(AB) = 0.$$

Now suppose A is invertible. Then A can be written as a product of elementary matrices:

$$A = E_1 \cdots E_\ell I.$$

Since each E_i is elementary,

$$\det(A) = \det(E_1) \cdots \det(E_\ell) \det(I) = \det(E_1) \cdots \det(E_\ell).$$

Next,

$$AB = (E_1 \cdots E_\ell)B,$$

so

$$\det(AB) = \det(E_1) \cdots \det(E_\ell) \det(B) = \det(A) \det(B).$$

☺

Corollary 10.1.6

If A is invertible, then

$$\det(A^{-1}) = \det(A)^{-1}.$$

$\det$ is our first structure-preserving map that respects matrix multiplication.

Theorem 10.1.7

The determinant is an invariant of similarity. If $A, B \in M_{n \times n}(F)$ are similar, i.e. if there exists an invertible matrix P such that

$$A = PBP^{-1},$$

then $\det(A) = \det(B)$.

Proof: Assume $A = PBP^{-1}$. Then

$$\det(A) = \det(P) \det(B) \det(P^{-1}) = \det(P) \det(B) \det(P)^{-1} = \det(B).$$

☺

Theorem 10.1.8

If $A \in M_{n \times n}(F)$ is diagonalizable, i.e. if there exists a basis $\{v_1, \dots, v_n\}$ of F^n such that $Av_i = \lambda_i v_i$ for eigenvalues $\lambda_1, \dots, \lambda_n$, then

$$\det(A) = \prod_{i=1}^n \lambda_i.$$

Proof: If A is diagonalizable, then there exists an invertible matrix P such that

$$A = PDP^{-1},$$

where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. By the previous theorem,

$$\det(A) = \det(D).$$

Since D is diagonal,

$$\det(D) = \prod_{i=1}^n \lambda_i.$$

**Theorem 10.1.9**

Let $A \in M_{n \times n}(F)$. The scalar λ is an eigenvalue of A iff

$$\det(A - \lambda I_n) = 0.$$

Proof: λ is an eigenvalue of A

$$\iff \exists v \in F^n \setminus \{0\} \text{ such that } Av = \lambda v$$

$$\iff Av - \lambda v = 0$$

$$\iff (A - \lambda I_n)v = 0$$

$$\iff A - \lambda I_n \text{ is not invertible}$$

$$\iff \det(A - \lambda I_n) = 0.$$

**Theorem 10.1.10**

$$\det(A) = \det(A^T)$$

Proof: From HW9, A invertible $\iff A^T$ invertible, so

$$\det(A) = 0 \iff \det(A^T) = 0.$$

If $\det(A) \neq 0$, then A is invertible, so

$$A = E_1 \cdots E_\ell$$

for elementary matrices E_i . Then (HW9)

$$A^T = E_\ell^T \cdots E_1^T.$$

Again by HW9, each E_i^T is an elementary matrix of the same type as E_i (with the same diagonal value c if E_i is type 2).

Thus

$$\det(A^T) = \prod_{i=\ell}^1 \det(E_i^T) = \prod_{i=1}^{\ell} \det(E_i) = \det(A).$$

☺

Corollary 10.1.7 Determinant via column expansion

For any fixed column j ,

$$\det(A) = \sum_{i=1}^n (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij}).$$

Proof: Fix a column index j . Consider the transpose A^T . By the theorem $\det(A) = \det(A^T)$, we have

$$\det(A) = \det(A^T).$$

Now apply the cofactor *row* expansion to A^T along its j -th row:

$$\det(A^T) = \sum_{i=1}^n (-1)^{j+i} (A^T)_{ji} \det(\widetilde{(A^T)}_{ji}).$$

But

$$(A^T)_{ji} = A_{ij},$$

and the minor $\widetilde{(A^T)}_{ji}$ is just the transpose of the minor $\tilde{A}_{ij}$, so

$$\det(\widetilde{(A^T)}_{ji}) = \det((\tilde{A}_{ij})^T) = \det(\tilde{A}_{ij}).$$

Thus

$$\det(A) = \sum_{i=1}^n (-1)^{j+i} A_{ij} \det(\tilde{A}_{ij}) = \sum_{i=1}^n (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij}),$$

which is the desired column expansion formula.

☺

Chapter 11

Week 11

11.1 Diagonalizability

Theorem 11.1.1 Eigenvalue criterion

Let $A \in M_{n \times n}(\mathbb{F})$. A scalar λ is an eigenvalue of A if and only if

$$\det(A - \lambda I_n) = 0.$$

Proof:

$$\lambda \text{ is an eigenvalue of } A \iff \exists 0 \neq v \text{ such that } Av = \lambda v.$$

$$\iff \exists 0 \neq v \text{ such that } Av = (\lambda I)v.$$

$$\iff \exists 0 \neq v \text{ such that } (A - \lambda I)v = 0.$$

$$\iff \ker(A - \lambda I) \neq 0.$$

$$\iff \det(A - \lambda I) = 0.$$



The last step is because if kernel is not equal to 0, then the matrix is not invertible, and thus the determinant is 0.

Example 11.1.1

$$\det \begin{pmatrix} 1-x & 0 & -1 \\ 3 & -6-x & 11 \\ 1 & -3 & 6-x \end{pmatrix} = (1-x) \det \begin{pmatrix} -6-x & 11 \\ -3 & 6-x \end{pmatrix} - 1 \cdot \det \begin{pmatrix} 3 & -6-x \\ 1 & -3 \end{pmatrix}.$$

Compute each 2×2 determinant:

$$\det \begin{pmatrix} -6-x & 11 \\ -3 & 6-x \end{pmatrix} = (-6-x)(6-x) - (11)(-3) = x^2 - 36 + 33 = x^2 - 3.$$

$$\det \begin{pmatrix} 3 & -6-x \\ 1 & -3 \end{pmatrix} = 3(-3) - (1)(-6-x) = -9 + 6 + x = x - 3.$$

Now substitute:

$$\det(A - xI) = (1-x)(x^2 - 3) - (x - 3).$$

Expand:

$$(1-x)(x^2 - 3) = x^2 - 3 - x^3 + 3x = -x^3 + x^2 + 3x - 3.$$

Subtract $x - 3$:

$$-x^3 + x^2 + 3x - 3 - (x - 3) = -x^3 + x^2 + 2x.$$

Factor:

$$-x^3 + x^2 + 2x = -x(x^2 - x - 2) = -x(x - 2)(x + 1).$$

$$\boxed{\det(A - xI) = -x(x - 2)(x + 1)}.$$

Definition 11.1.1: Characteristic Polynomial

Let $A \in M_{n \times n}(\mathbb{F})$. The polynomial

$$f(t) = \det(A - tI_n)$$

is called the *characteristic polynomial* of A .

Let V be a finite-dimensional vector space and let $T : V \rightarrow V$ be linear. The characteristic polynomial of T is defined to be the characteristic polynomial of the matrix $[T]_\beta$ for any basis β of V .

Example 11.1.2 (Basis-independence of the characteristic polynomial)

Check that the characteristic polynomial does not depend on the choice of basis. Another way to restate our first theorem is: the eigenvalues of A are precisely the roots of $f_A(t)$.

Proposition 11.1.1 Degree of the characteristic polynomial

Let $A \in M_{n \times n}(\mathbb{F})$. Then the characteristic polynomial

$$f_A(t) = \det(A - tI)$$

is a polynomial of degree n .

Proof: We proceed by induction on n .

For $n = 1$, we have

$$f_A(t) = \det(a_{11} - t) = a_{11} - t,$$

which is a polynomial of degree 1.

Assume the result holds for all $(n - 1) \times (n - 1)$ matrices. Recall

$$f_A(t) = \det(A - tI).$$

Write the matrix

$$A - tI = \begin{pmatrix} a_{11} - t & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - t & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - t \end{pmatrix}.$$

Expanding along the first row gives:

$$\det(A - tI) = (a_{11} - t) \det(\tilde{A}_{11} - tI_{n-1}) + \sum_{i=2}^n (-1)^{i+1} a_{1i} \det(\tilde{A}_{1i} - tI_{n-1}),$$

where $\tilde{A}_{1i}$ denotes the $(n - 1) \times (n - 1)$ minor.

By the induction hypothesis,

$$\det(\tilde{A}_{11} - tI_{n-1})$$

is a polynomial of degree $n - 1$, so the first term

$$(a_{11} - t) \det(\tilde{A}_{11} - tI_{n-1})$$

has degree n .

For $i \geq 2$, the minors $\tilde{A}_{1i}$ have a first row which does not contain t . Thus when expanding $\det(\tilde{A}_{1i} - tI)$ along that row, all resulting minors are polynomials of degree at most $n - 2$.

Therefore, each term

$$a_{1i} \det(\tilde{A}_{1i} - tI) \quad (i \geq 2)$$

has degree at most $n - 2$.

Putting everything together, the sum

$$\det(A - tI)$$

has highest degree n .

Thus $f_A(t)$ is a polynomial of degree n .



Theorem 11.1.2 Root-factor characterization of polynomials

Let $p(t)$ be a polynomial of degree n . Then λ is a root of $p(t)$ if and only if

$$p(t) = (t - \lambda) q(t)$$

for some polynomial $q(t)$ of degree $n - 1$.

In particular, a polynomial of degree n has at most n roots.

Proof. Standard; see MAT247.

Corollary 11.1.1 Eigenvalue bound

A matrix $A \in M_{n \times n}(\mathbb{F})$ has at most n eigenvalues.

Definition 11.1.2: Algebraic multiplicity

Let λ be an eigenvalue of A . The *algebraic multiplicity* of λ is the largest integer k such that

$$f_A(t) = (t - \lambda)^k q(t)$$

for some polynomial $q(t)$ of degree $n - k$. In other words, it is the number of times $(t - \lambda)$ appear as a factor of $f_A(t)$.

Theorem 11.1.3 Eigenvectors from kernels

Let $A \in M_{n \times n}(\mathbb{F})$, and let λ be an eigenvalue. Then v is an eigenvector with eigenvalue λ if and only if $v \neq 0$ and

$$v \in \ker(A - \lambda I).$$

Proof: Suppose first that v is an eigenvector of A with eigenvalue λ . By definition this means

$$Av = \lambda v.$$

Rewriting,

$$(A - \lambda I)v = 0.$$

Thus $v \in \ker(A - \lambda I)$, and of course $v \neq 0$ because eigenvectors are required to be nonzero.

Conversely, suppose $v \neq 0$ and

$$v \in \ker(A - \lambda I).$$

Then

$$(A - \lambda I)v = 0 \implies Av = \lambda v.$$

Hence v is an eigenvector of A with eigenvalue λ .

Therefore v is an eigenvector with eigenvalue λ if and only if $v \neq 0$ and $v \in \ker(A - \lambda I)$. $\odot$

Example 11.1.3 (Eigenvalues and eigenvectors of a 3×3 matrix)

Let

$$A = \begin{pmatrix} 14 & 0 & 5 \\ -4 & 1 & -2 \\ -30 & 0 & -11 \end{pmatrix}.$$

First compute the characteristic polynomial. One verifies that

$$f_A(t) = (4 - t)(1 - t)(1 + t).$$

To find the eigenvectors associated to $\lambda = 4$, compute

$$\ker(A - 4I) = \ker \begin{pmatrix} 10 & 0 & 5 \\ -4 & -3 & -2 \\ -30 & 0 & -15 \end{pmatrix}.$$

A vector in the kernel is a linear relation among the columns. Here

$$c_1 - 2c_3 = 0,$$

so an eigenvector is

$$v_4 = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}.$$

Next consider $\lambda = 1$:

$$\ker(A - I) = \ker \begin{pmatrix} 13 & 0 & 5 \\ -4 & 0 & -2 \\ -30 & 0 & -12 \end{pmatrix}.$$

This yields

$$\ker(A - I) = \text{span} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Next consider $\lambda = -1$:

$$\ker(A + I) = \ker \begin{pmatrix} 15 & 0 & 5 \\ -4 & 2 & -2 \\ -30 & 0 & -10 \end{pmatrix}.$$

Thus

$$\ker(A + I) = \text{span} \begin{pmatrix} 1 \\ -1 \\ -3 \end{pmatrix}.$$

The eigenvalues of A are therefore

$$4, \quad 1, \quad -1,$$

with eigenvectors

$$\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ -1 \\ -3 \end{pmatrix}.$$

Since A has three distinct eigenvalues, it is diagonalizable. Let

$$P = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ -2 & 0 & -3 \end{pmatrix}, \quad D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Then

$$A = PDP^{-1}.$$

Definition 11.1.3: Linear operator

A linear operator $T : V \rightarrow V$ is a linear transformation from a vector space to itself.

Theorem 11.1.4 Linear independence of eigenvectors

Let T be a linear operator on a vector space, and let

$$\lambda_1, \dots, \lambda_k$$

be distinct eigenvalues of T . For each i , let S_i be a finite set of eigenvectors of T with eigenvalue λ_i . If each S_i is linearly independent, then the union

$$S_1 \cup \dots \cup S_k$$

is linearly independent.

Proof: We proceed by induction on k .

If $k = 1$, there is nothing to prove. Assume the statement holds for $k - 1$, that is,

$$S_1 \cup \dots \cup S_{k-1} \quad \text{is linearly independent.}$$

Write

$$S_i = \{v_{i1}, \dots, v_{i,|S_i|}\}.$$

Assume there is a linear relation

$$\sum_{i=1}^k \sum_{j=1}^{|S_i|} a_{ij} v_{ij} = 0.$$

Apply the operator $T - \lambda_k I$ to both sides. Using that $Tv_{ij} = \lambda_i v_{ij}$, we get

$$0 = \sum_{i=1}^k \sum_{j=1}^{|S_i|} a_{ij}(\lambda_i - \lambda_k)v_{ij}.$$

Since $\lambda_k - \lambda_k = 0$, the terms with $i = k$ vanish, so

$$0 = \sum_{i=1}^{k-1} \sum_{j=1}^{|S_i|} a_{ij}(\lambda_i - \lambda_k)v_{ij}.$$

Because the eigenvalues $\lambda_1, \dots, \lambda_{k-1}$ are distinct from λ_k , each factor $(\lambda_i - \lambda_k)$ is nonzero. Hence

$$\sum_{i=1}^{k-1} \sum_{j=1}^{|S_i|} a_{ij}v_{ij} = 0.$$

By the induction hypothesis, the vectors in

$$S_1 \cup \dots \cup S_{k-1}$$

are linearly independent, so

$$a_{ij} = 0 \quad (1 \leq i \leq k-1, 1 \leq j \leq |S_i|).$$

The original relation now reduces to

$$\sum_{j=1}^{|S_k|} a_{kj}v_{kj} = 0.$$

Since S_k is linearly independent, all coefficients a_{kj} must vanish. Therefore every $a_{ij} = 0$, and the only linear relation among the vectors in

$$S_1 \cup \dots \cup S_k$$

is the trivial one.

Thus $S_1 \cup \dots \cup S_k$ is linearly independent. ⊙

Corollary 11.1.2 Distinct eigenvalues imply diagonalizable

Let T be a linear operator on an n -dimensional vector space V with n distinct eigenvalues. Then T is diagonalizable.

Proof: Apply the previous theorem with $S_i = \{v_i\}$, where v_i is an eigenvector with eigenvalue λ_i . Since the eigenvalues are all distinct, each S_i is linearly independent, and the theorem implies that the union

$$\{v_1, \dots, v_n\}$$

is linearly independent. Thus this set is a basis of V consisting of eigenvectors of T , so T is diagonalizable. ⊙

Definition 11.1.4: Splitting of a polynomial

Let $p(t) \in \mathbb{F}[t]$ be a polynomial of degree n . We say that $p(t)$ *splits over* $\mathbb{F}$ if $p(t)$ has n roots in $\mathbb{F}$; i.e.,

$$p(t) = c \prod_{i=1}^n (t - \lambda_i) \quad \text{for some } c \in \mathbb{F}.$$

For example, the polynomial $(x - 1)(x^2 + 1)$ does not split over $\mathbb{R}$.

Theorem 11.1.5

If $A \in M_{n \times n}(\mathbb{F})$ is diagonalizable, then its characteristic polynomial $f_A(t)$ splits over $\mathbb{F}$.

Proof: Suppose $A = PDP^{-1}$ with $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then

$$f_A(t) = \det(A - tI) = \det(D - tI) = (\lambda_1 - t) \cdots (\lambda_n - t),$$

which is a product of linear factors in $\mathbb{F}[t]$. Thus $f_A(t)$ splits over $\mathbb{F}$. $\odot$

Definition 11.1.5: Eigenspace

Let $\lambda \in F$ be an eigenvalue of a linear operator $T : V \rightarrow V$. The eigenspace of T associated to λ is the subset

$$E_\lambda = \{v \in V \mid Tv = \lambda v\}.$$

Question 1

Check that E_λ is a subspace of V .

Proposition 11.1.2 Dimension of Eigenspace

Let $T : V \rightarrow V$ be linear and let λ be an eigenvalue of T . Then $\dim(E_\lambda) \geq 1$.

Proof: Since λ is an eigenvalue, by definition there exists a nonzero vector v such that $Tv = \lambda v$. Thus $v \in E_\lambda$, so E_λ contains a nonzero vector. Hence $\dim(E_\lambda) \geq 1$. $\odot$

Theorem 11.1.6 Diagonalizability Criterion

Let T be a linear operator on a vector space V with $\dim(V) = n$, and suppose the characteristic polynomial $f_T(t)$ splits over the field. Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T .

Then T is diagonalizable if and only if

$$m_i = \dim(E_{\lambda_i}) \quad \text{for all } i,$$

where m_i is the algebraic multiplicity of λ_i in $f_T(t)$.

Moreover, if T is diagonalizable and β_i is an ordered basis of E_{λ_i} , then the concatenation

$$\beta = \beta_1 \cup \cdots \cup \beta_k$$

is an ordered basis of V consisting entirely of eigenvectors of T .

Proof: For each i , let m_i be the multiplicity of the eigenvalue λ_i , and let $d_i = \dim(E_{\lambda_i})$.

($\Rightarrow$) **Assume T is diagonalizable.** Let β be a basis of V consisting of eigenvectors of T . For each i , define

$$\beta_i = \beta \cap E_{\lambda_i}.$$

Then β_i is linearly independent, so

$$|\beta_i| \leq d_i.$$

On the other hand, by the previous theorem, all eigenvectors corresponding to distinct eigenvalues are linearly independent. Since every vector in β lies in exactly one E_{λ_i} , we have

$$n = \sum_i |\beta_i|.$$

But also $|\beta_i| \leq d_i$ for each i , so

$$n = \sum_i |\beta_i| \leq \sum_i d_i \leq \sum_i m_i = n.$$

Hence all inequalities must be equalities, implying

$$m_i = d_i \quad \text{for all } i.$$

($\Leftarrow$) **Suppose $m_i = d_i$ for all i .** For each i , let β_i be a basis of E_{λ_i} , and define

$$\beta = \beta_1 \cup \cdots \cup \beta_k.$$

Since eigenvectors corresponding to distinct eigenvalues are linearly independent, β is linearly independent.

Moreover,

$$|\beta| = \sum_i |\beta_i| = \sum_i d_i = \sum_i m_i = n.$$

Thus β is a basis of V , and every vector in β is an eigenvector. Therefore T is diagonalizable. $\odot$

Theorem 11.1.7 Diagonalizable $\Rightarrow$ Characteristic Polynomial Splits

Let $A \in M_n(F)$. If A is diagonalizable over F , then its characteristic polynomial

$$f_A(t) = \det(tI - A)$$

splits over F .

Proof: Assume $A = PDP^{-1}$ for some invertible matrix P , where

$$D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

is diagonal. Then

$$\begin{aligned}
 f_A(t) &= \det(tI - A) = \det(tI - PDP^{-1}) \\
 &= \det(P(tI - D)P^{-1}) \\
 &= \det(P) \det(tI - D) \det(P^{-1}) \\
 &= \det(P) \det(tI - D) \frac{1}{\det(P)} \\
 &= \det(tI - D)
 \end{aligned}$$

But

$$\det(tI - D) = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_n),$$

which shows that the characteristic polynomial splits completely over F . $\odot$

Example 11.1.4 (Eigenspace is a Subspace)

Show that E_λ is a subspace of V .

Proof: Let $u, v \in E_\lambda$. Then

$$T(u) = \lambda u, \quad T(v) = \lambda v.$$

For any scalars $a, b \in F$,

$$T(au + bv) = aT(u) + bT(v) = a(\lambda u) + b(\lambda v) = \lambda(au + bv).$$

Thus $au + bv \in E_\lambda$.

Also, $0 \in E_\lambda$ since $T(0) = 0 = \lambda \cdot 0$. Hence E_λ is a subspace of V . $\odot$

Theorem 11.1.8

Let $T : V \rightarrow V$ be a linear operator with $\dim V = n$, and let λ be an eigenvalue of T with algebraic multiplicity k . Then

$$1 \leq \dim(E_\lambda) \leq k.$$

Proof: Choose a basis $\{v_1, \dots, v_p\}$ of the eigenspace E_λ , and extend it to a basis

$$\beta = \{v_1, \dots, v_p, v_{p+1}, \dots, v_n\}$$

of V .

Let $A = [T]_\beta$. Since $T(v_i) = \lambda v_i$ for $1 \leq i \leq p$, the matrix A has the block form

$$A = \begin{pmatrix} \lambda I_p & B \\ 0 & C \end{pmatrix},$$

where $B \in M_{p \times (n-p)}(F)$ and $C \in M_{(n-p) \times (n-p)}(F)$.

Now compute the characteristic polynomial:

$$\begin{aligned}
 f_A(t) &= \det(A - tI_n) \\
 &= \det \begin{pmatrix} (\lambda - t)I_p & B \\ 0 & C - tI_{n-p} \end{pmatrix} \\
 &= \det((\lambda - t)I_p) \det(C - tI_{n-p}) \\
 &= (\lambda - t)^p f_C(t).
 \end{aligned}$$

Thus $(\lambda - t)^p$ divides $f_A(t)$, so the algebraic multiplicity k of λ satisfies $p \leq k$. Since $p = \dim(E_\lambda)$, we obtain

$$\dim(E_\lambda) \leq k.$$

The inequality $\dim(E_\lambda) \geq 1$ holds because λ is an eigenvalue. ☺

Example 11.1.5

Compute the characteristic polynomial and determine the dimension of the eigenspace for

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Characteristic polynomial.

$$f_A(t) = \det(A - tI) = \det \begin{pmatrix} 1-t & 1 \\ 0 & 1-t \end{pmatrix} = (1-t)(1-t) - 0 \cdot 1 = (1-t)^2.$$

So the only eigenvalue is $\lambda = 1$, with algebraic multiplicity 2.

Eigenspace and its dimension.

$$E_1 = \ker(A - I).$$

Compute

$$A - I = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Solve $(A - I) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$:

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies y = 0,$$

with x free. Hence

$$E_1 = \left\{ \begin{pmatrix} x \\ 0 \end{pmatrix} : x \in F \right\} = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}, \quad \dim(E_1) = 1.$$

Conclusion.

$$f_A(t) = (1 - t)^2, \quad \lambda = 1, \quad \dim(E_1) = 1.$$

In particular, the geometric multiplicity 1 is strictly less than the algebraic multiplicity 2, so A is not diagonalizable.

Theorem 11.1.9 Diagonalizability via Eigenspaces

Let $T : V \rightarrow V$ be a linear operator on a vector space V with $\dim(V) = n$. Assume that the characteristic polynomial $f_T(t)$ splits over the field F , and let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T .

Then T is diagonalizable if and only if the algebraic multiplicity of λ_i equals $\dim(E_{\lambda_i})$ for all $i = 1, \dots, k$.

Proof: For each i , let m_i denote the algebraic multiplicity of λ_i , and let $d_i = \dim(E_{\lambda_i})$.

($\Rightarrow$) Assume that T is diagonalizable. Then there exists a basis β of V consisting entirely of eigenvectors of T . For each i , define

$$\beta_i = \beta \cap E_{\lambda_i}.$$

Then β_i is a linearly independent subset of E_{λ_i} , hence

$$|\beta_i| \leq d_i.$$

On the other hand, by the previous theorem, we know that $d_i \leq m_i$ for all i . Since every vector in β belongs to exactly one β_i , we have

$$n = \sum_i |\beta_i| \leq \sum_i d_i \leq \sum_i m_i = n.$$

It follows that all inequalities are equalities, hence $|\beta_i| = d_i = m_i$ for all i .

($\Leftarrow$) Conversely, suppose that $m_i = d_i$ for all i . For each i , let β_i be a basis of E_{λ_i} , and define

$$\beta = \beta_1 \cup \dots \cup \beta_k.$$

Since eigenspaces corresponding to distinct eigenvalues intersect trivially, the set β is linearly independent. Moreover,

$$|\beta| = \sum_i |\beta_i| = \sum_i d_i = \sum_i m_i = n,$$

so β is a basis of V consisting entirely of eigenvectors.

Therefore, T is diagonalizable. ☺

Example 11.1.6 (Diagonalizability Examples)

Example 1. Is

$$A = \begin{pmatrix} 2 & -10 & -1 \\ 0 & -3 & 0 \\ 1 & -2 & 0 \end{pmatrix}$$

diagonalizable?

Compute the characteristic polynomial:

$$f_A(t) = \det(A - tI) = \det \begin{pmatrix} 2-t & -10 & -1 \\ 0 & -3-t & 0 \\ 1 & -2 & -t \end{pmatrix}.$$

Expanding along the second row,

$$f_A(t) = (-3-t) \det \begin{pmatrix} 2-t & -1 \\ 1 & -t \end{pmatrix}.$$

Thus

$$f_A(t) = (-3-t)((2-t)(-t) + 1) = (-3-t)(t^2 - 2t + 1) = (-3-t)(t-1)^2.$$

Hence the eigenvalues are -3 and 1 , with algebraic multiplicities

$$m(-3) = 1, \quad m(1) = 2.$$

To test diagonalizability, compute the eigenspace for $\lambda = -3$:

$$E_{-3} = \ker(A + 3I) = \ker \begin{pmatrix} 5 & -10 & -1 \\ 0 & 0 & 0 \\ 1 & -2 & 3 \end{pmatrix}.$$

The rank of this matrix is 2, so the nullity is 1. Therefore

$$\dim E_{-3} = 1.$$

Since the algebraic multiplicity of -3 is 2 but

$$\dim E_{-3} = 1 < 2,$$

the matrix A is *not diagonalizable*.

Example 11.1.7 (Diagonalizability with Simple Eigenvalues)

Is

$$A = \begin{pmatrix} 2 & 0 & -1 \\ 8 & -6 & -1 \\ -2 & 2 & 1 \end{pmatrix}$$

diagonalizable?

Compute the characteristic polynomial:

$$f_A(t) = \det(A - tI) = \det \begin{pmatrix} 2-t & 0 & -1 \\ 8 & -6-t & -1 \\ -2 & 2 & 1-t \end{pmatrix}.$$

Expanding,

$$\begin{aligned} f_A(t) &= (2-t)((-6-t)(1-t) - (-1)(2)) - (8 \cdot 2 - (-2)(-6-t)) \\ &= (2-t)(t^2 + 5t - 6). \end{aligned}$$

Factoring,

$$f_A(t) = (2-t)(t+6)(t-1).$$

Thus the eigenvalues are $2, -6, 1$, each with algebraic multiplicity 1.

Since all eigenvalues are distinct, each eigenspace has dimension 1, and therefore A is diagonalizable.